# THE UNIVERSITY OF ZAMBIA

## School of Natural Sciences

### Department of Computing and Informatics

**4th Year Project Presentation**

**16 October 2026**

# COMPUTER SCIENCE
# 4TH YEAR PROJECT PRESENTATION

***Student Information System Design and Implementation with Moodle Integration***

**Chitundu Milimbo and Charles Hangoma**

**Supervisor: Professor Jackson Phiri**

**16 October 2026**

# PRESENTATION OUTLINE

- Introduction and background
- Problem statement
- Aim and objectives
- Research questions
- Literature review
- Methodology and system design
- Results and discussion
- Conclusion and recommendations
- References

# INTRODUCTION

- We developed a Student Information System (SIS) for university administration and student services.
- Moodle supports teaching and learning. Our SIS owns registration and official academic records.
- Our project connects these responsibilities through controlled integration workflows.

# BACKGROUND OF THE STUDY

- Students depend on consistent records across admissions, registration and their learning platform.
- Moodle web services support external access to users, courses and enrolments.
- We use a configurable university structure and demonstration policies to exercise these workflows.

# STATEMENT OF THE PROBLEM

- Disconnected administrative and learning records create duplicate work and inconsistent course access.
- A successful registration must remain valid when Moodle is unavailable.
- Staff need traceable approval, reconciliation and recovery rather than manual edits without history.
- **Students need clear registration status, personal timetables and useful support routes.**

# AIM OF THE STUDY

- To design and implement a unified Student Information System with bidirectional Moodle integration and a governed AI layer for administrative workflows and proactive student support.

# RESEARCH OBJECTIVES

- Develop modular SIS administration and integrate Moodle using web services and LTI 1.3.
- Implement governed AI for student enquiries, staff summaries, risk insights and opt-in wellbeing support.

# RESEARCH QUESTIONS

How can we preserve authoritative SIS records while linking course access in Moodle?

- How can configurable policies support university workflows and independent approvals?
- What controls must precede AI assistance using student information?

# LITERATURE REVIEW (1/3)

- Moodle web services provide external functions for user, course and enrolment operations [1].
- We use this provisioning lane to demonstrate course access on MoodleCloud.
- Our design separates successful SIS registration from later integration delivery.

# LITERATURE REVIEW (2/3)

- LTI 1.3 defines secure launches between a learning platform and an external tool [2].
- This supports a separate tool launch lane alongside web service provisioning.
- LTI launch and single sign-on remain future implementation work in our current build.

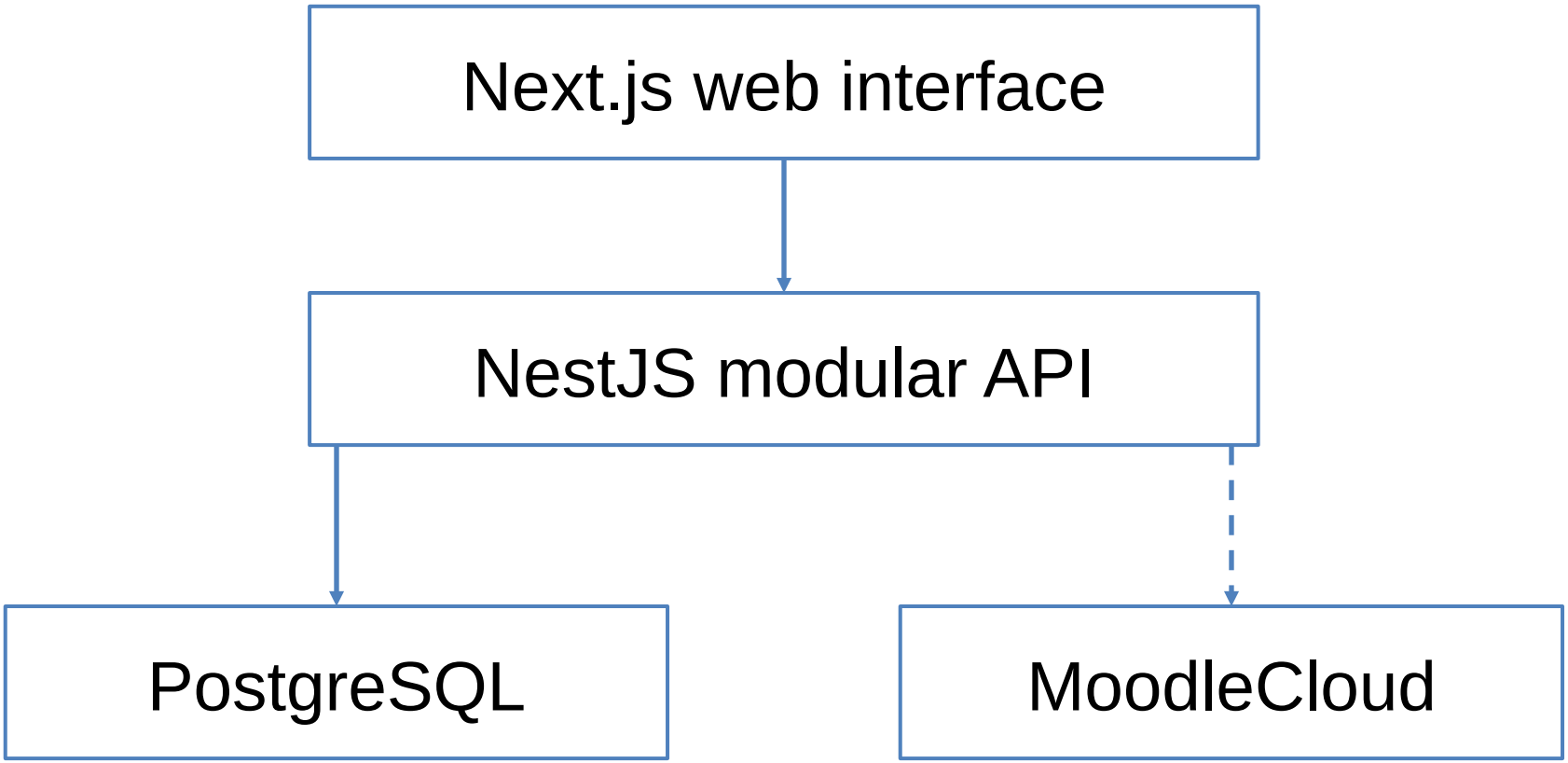
# LITERATURE REVIEW (3/3)

- NIST AI RMF 1.0 structures AI risk management around Govern, Map, Measure and Manage [3].
- Our proposed assistant requires permission checks, approved knowledge and human escalation.
- We have not completed or evaluated a deployed student AI assistant.

# RESEARCH METHODOLOGY (1/4)

- **Iterative system development**
  - **We map handbook requirements to task packets and vertical slices.**
  - **We implement UI, API, database state and audit behaviour together.**
  - **We test isolated policies and representative student workflows.**
  - **We record unresolved capabilities separately from demonstrated behaviour.**

# RESEARCH METHODOLOGY (2/4)




Dashed path: live delivery pending

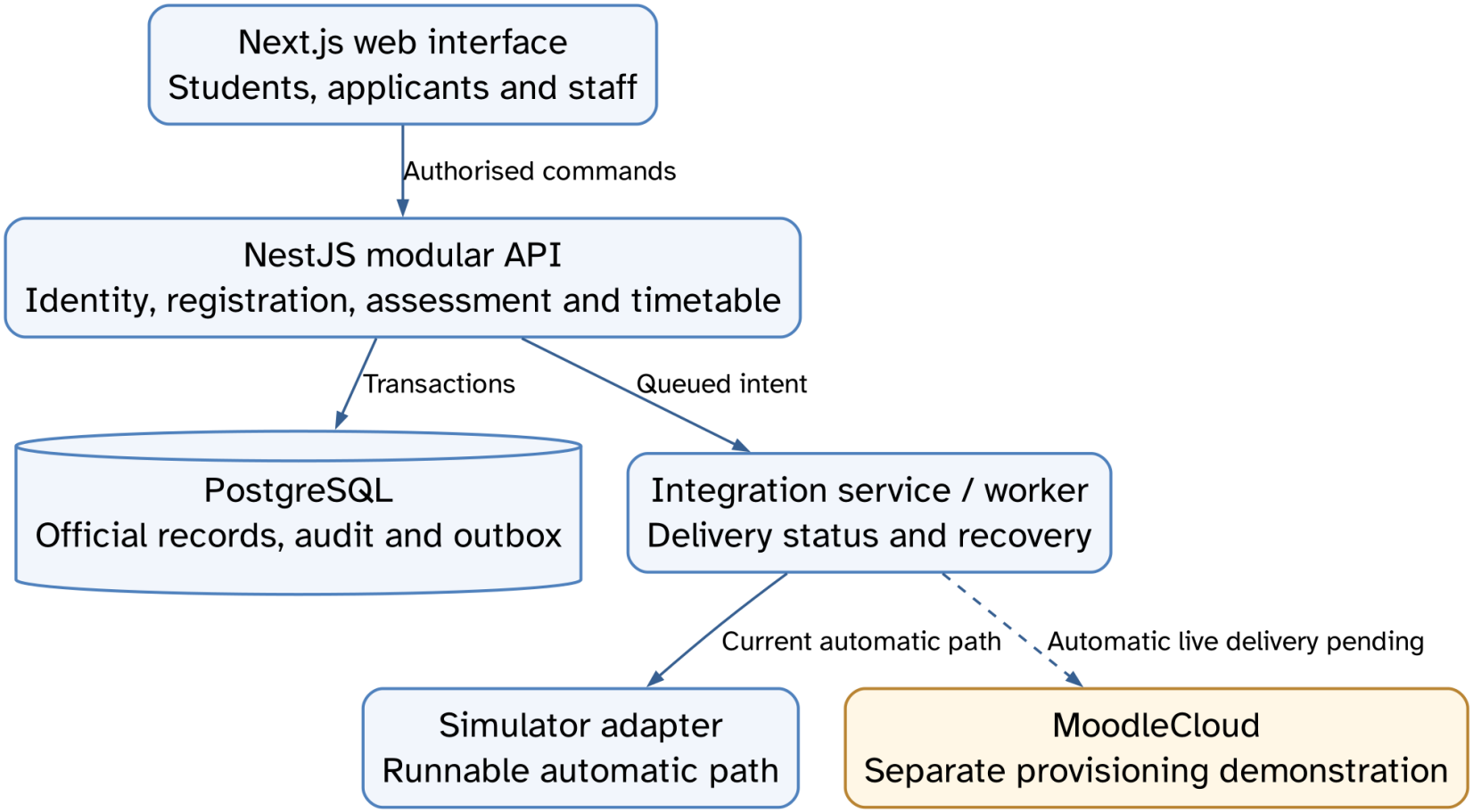
# RESEARCH METHODOLOGY (3/4)

- **PostgreSQL stores academic records, versioned decisions and integration evidence.**
- **Scoped roles control access to each workspace and student record.**
- **Independent reviewers approve sensitive publication decisions.**
- **Audit records and immutable snapshots preserve the history of consequential changes.**

# RESEARCH METHODOLOGY (4/4)

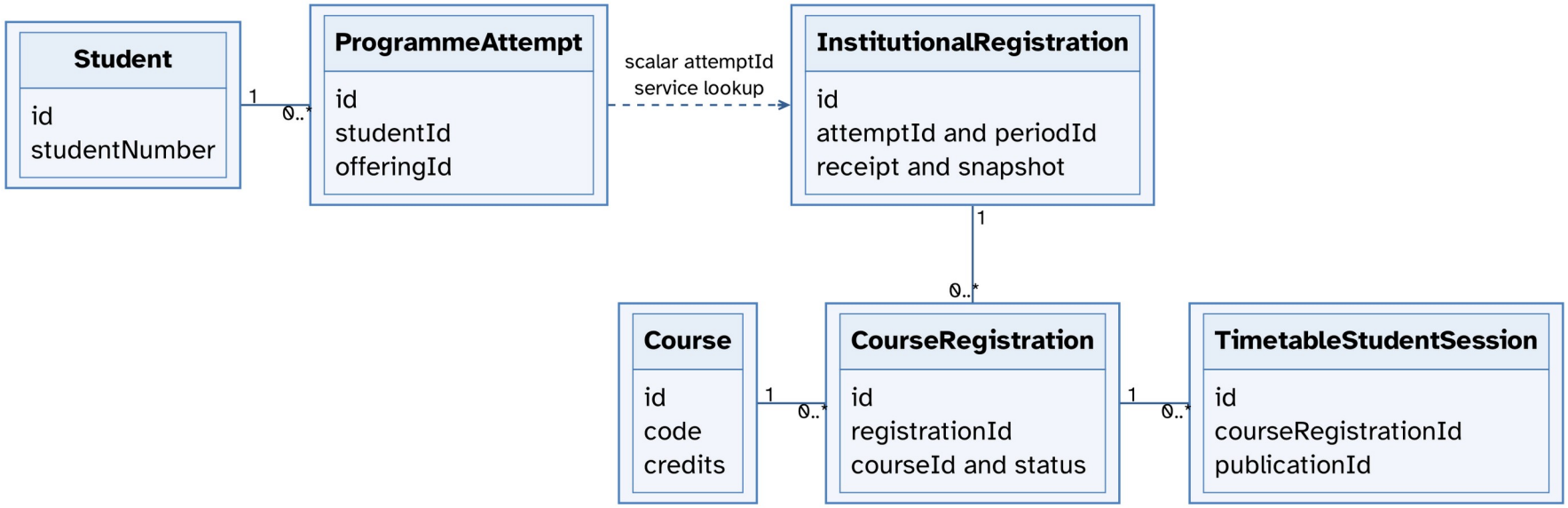
- We use API tests, connected database checks and browser journeys.
- Current local checks passed four sign-in journeys and six workspace database checks.
- A prior release passed 137 main browser tests and 26 Moodle deferral tests.
- These checks do not replace institutional user acceptance testing.

# SYSTEM ARCHITECTURE



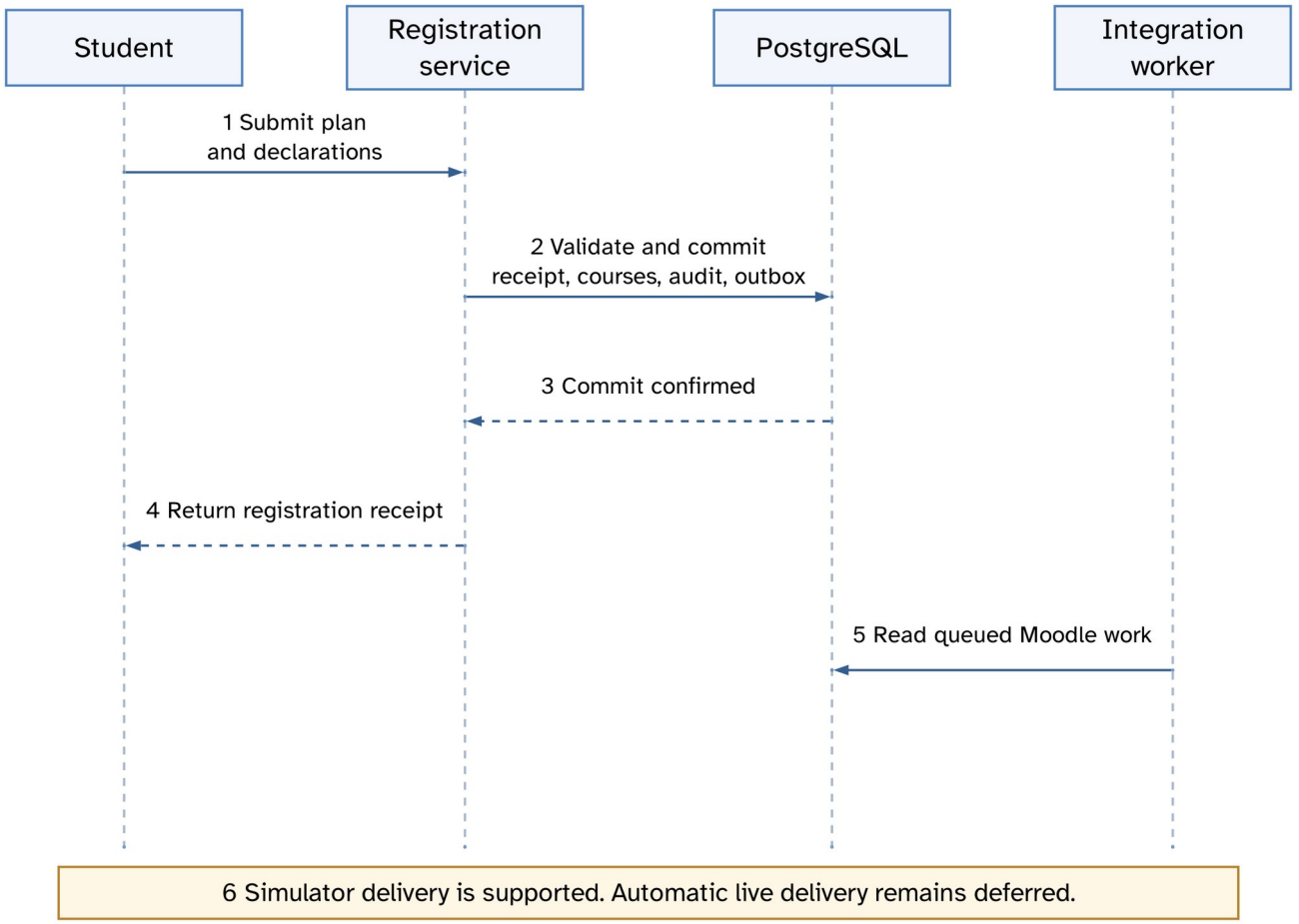

SIS decisions commit before external delivery.

# DOMAIN CLASS MODEL



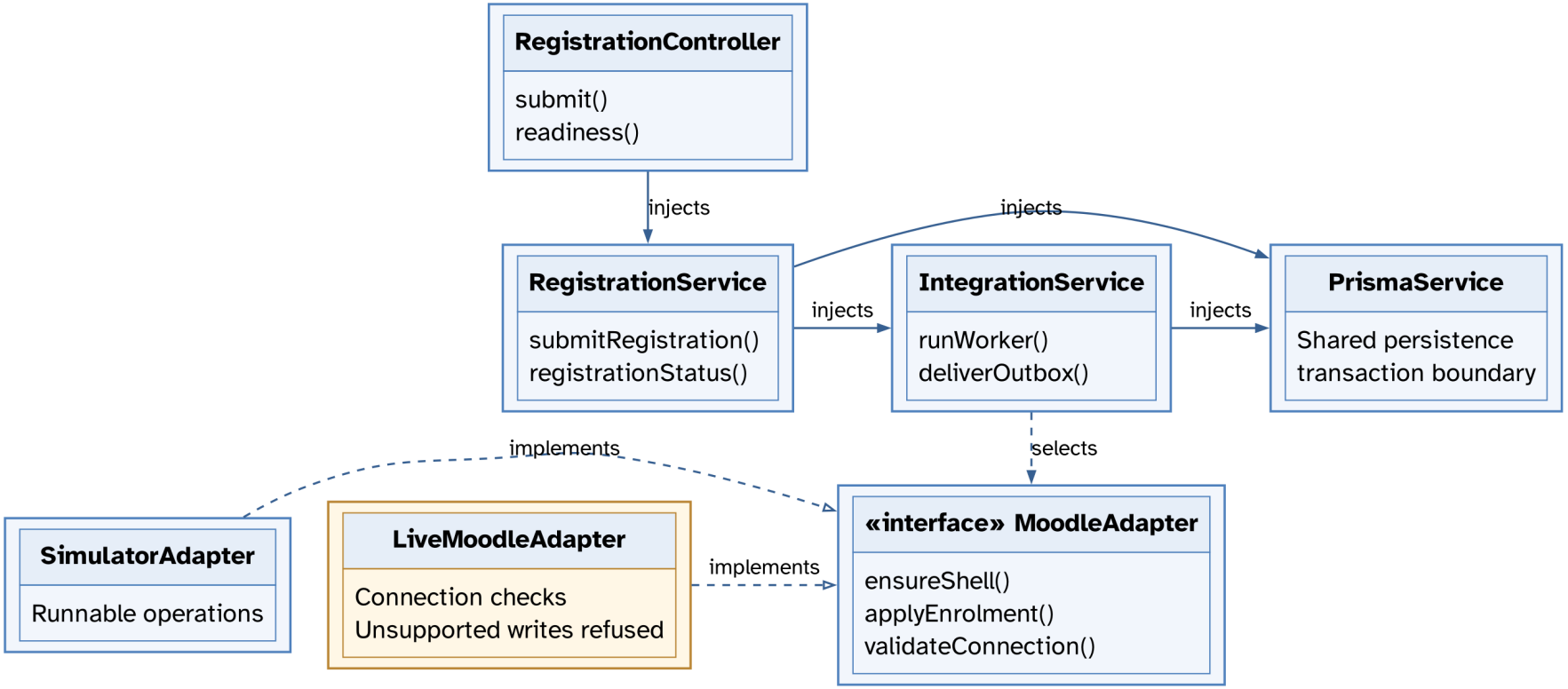

Solid: schema relation. Dashed: scalar-key lookup.

# REGISTRATION SEQUENCE



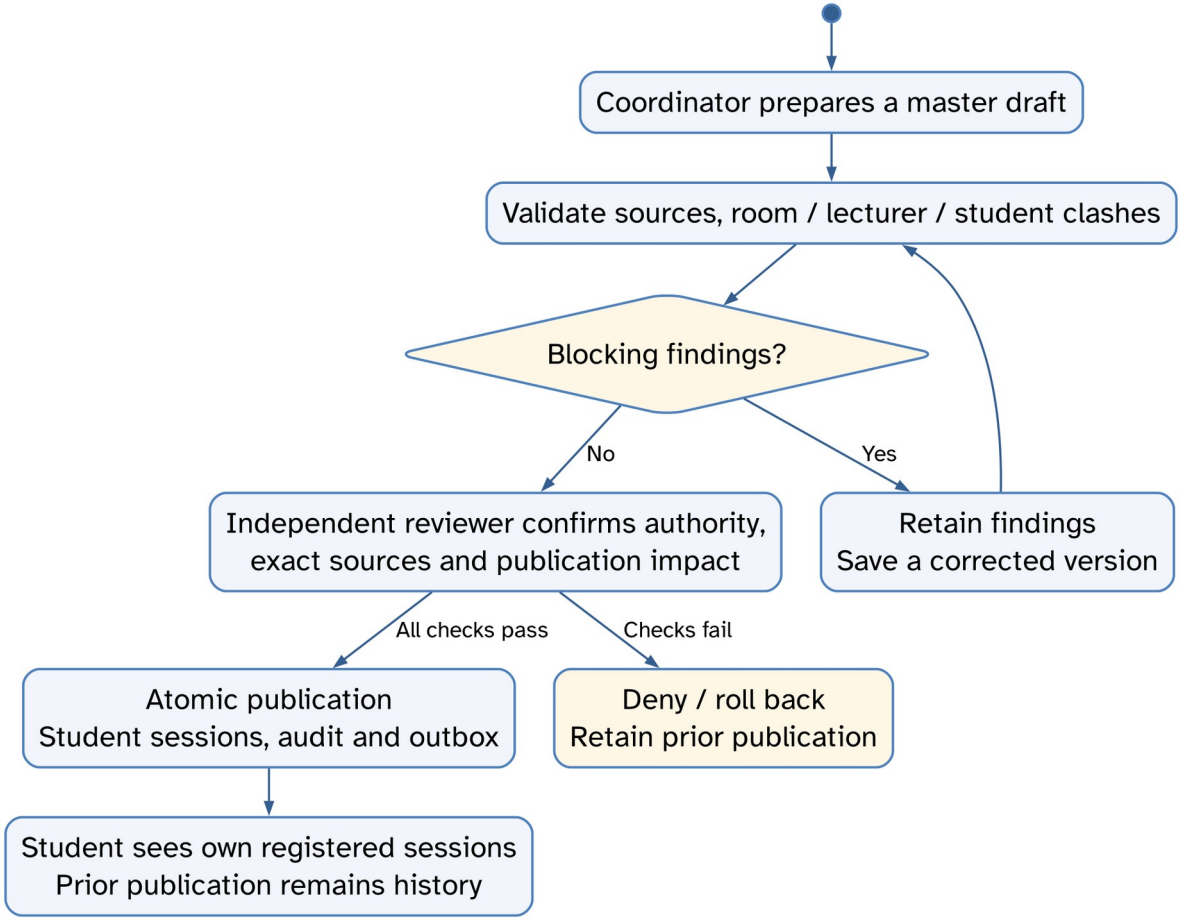

A Moodle delay does not undo a valid registration.

# DESIGN CLASS DIAGRAM



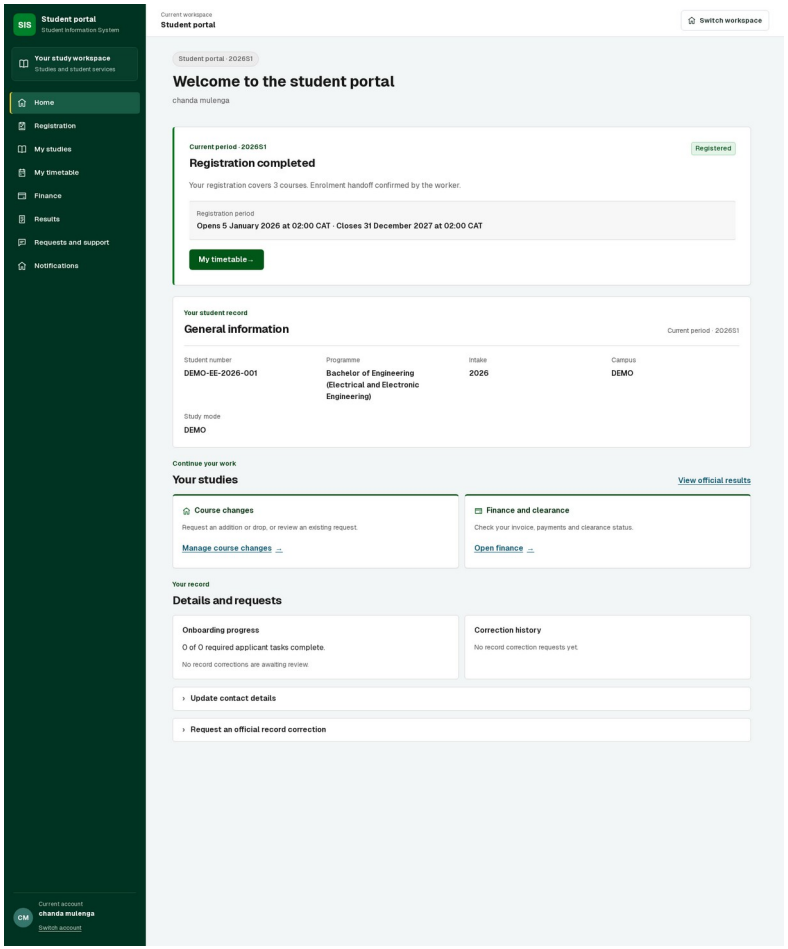

Controllers call services; adapters isolate delivery.

# TIMETABLE PUBLICATION



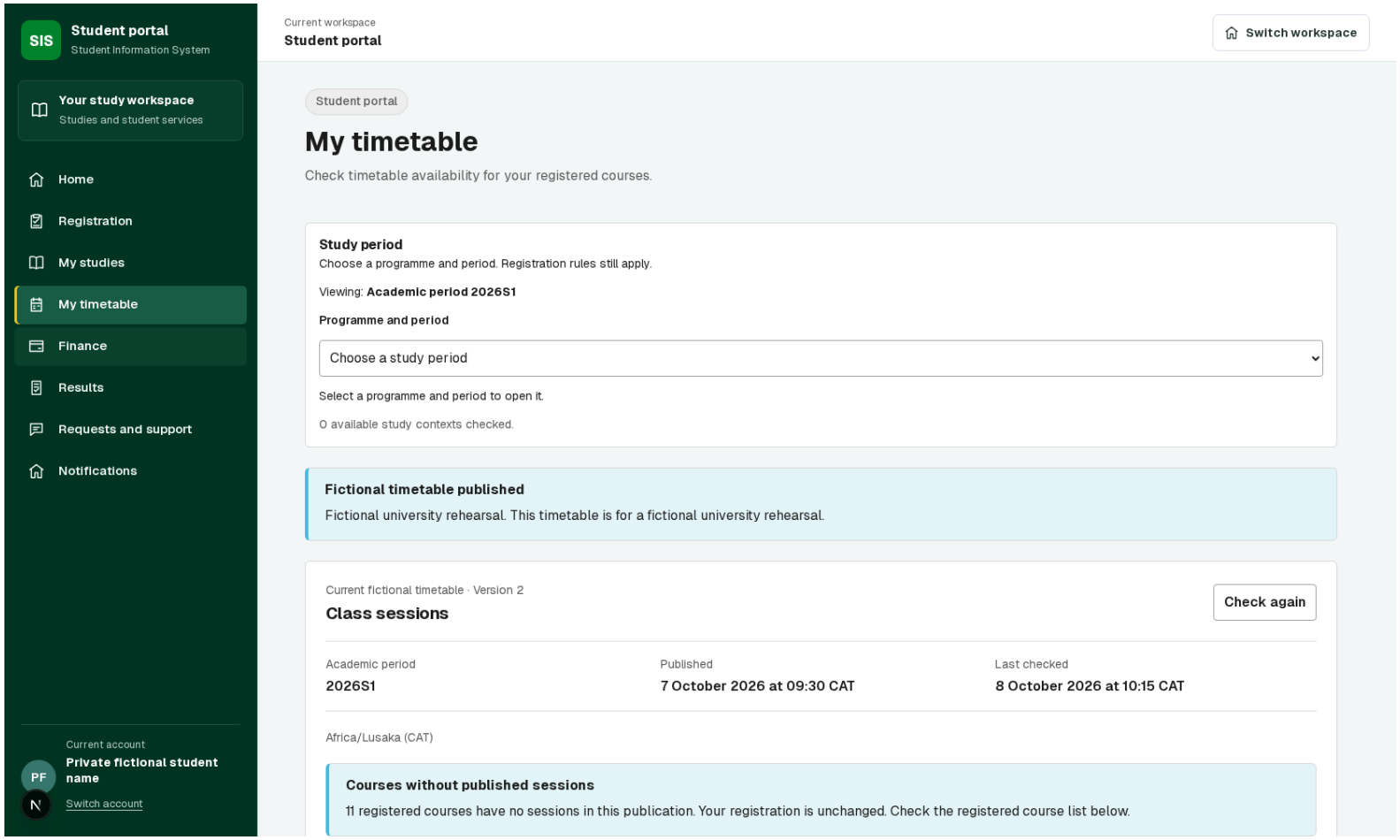

Independent review precedes publication.

# RESULTS AND DISCUSSION (1/4)




Complete capture. Enlarged page details follow the references.

# RESULTS AND DISCUSSION (2/4)



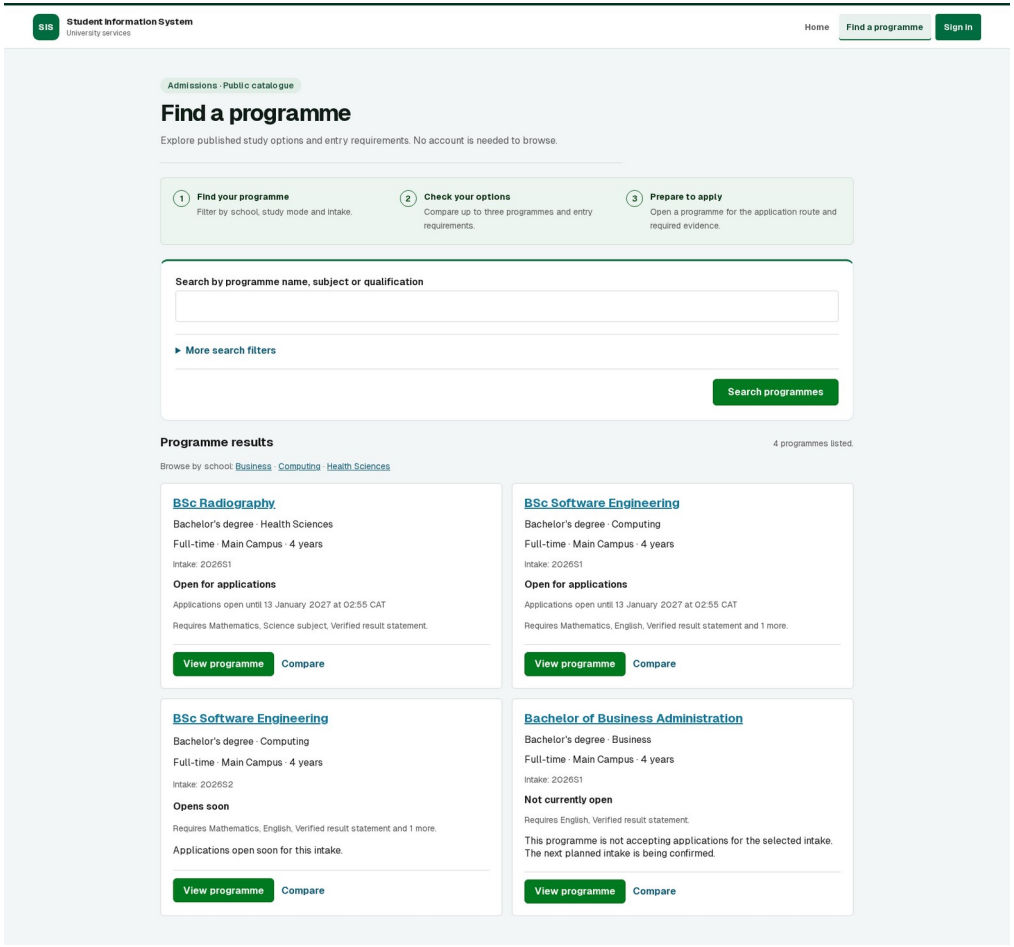
# RESULTS AND DISCUSSION (3/4)

- We created 12 visible MoodleCloud courses and four demonstration accounts.
- We verified 48 course memberships across those accounts.
- A learner signed in normally, saw all 12 courses and opened a course.
- Automatic live synchronisation and grade delivery still require completion.

# RESULTS AND DISCUSSION (4/4)

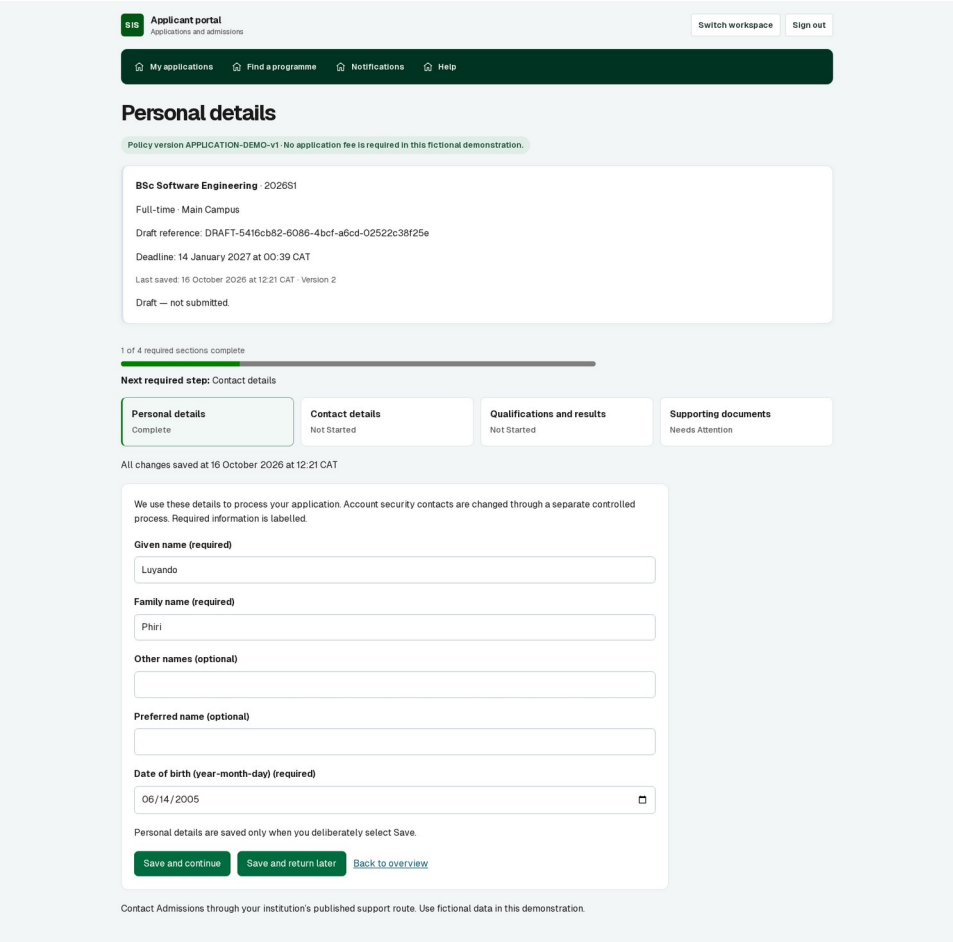
- Our prepared SIS dataset completed nine registrations through the API.
- Published timetables project relevant sessions to each registered student.
- Registration failures preserve actionable validation messages and supporting evidence.
- The latest student UI is a local checkpoint, separate from the earlier deployed release.

# 4.1 PROGRAMME CATALOGUE



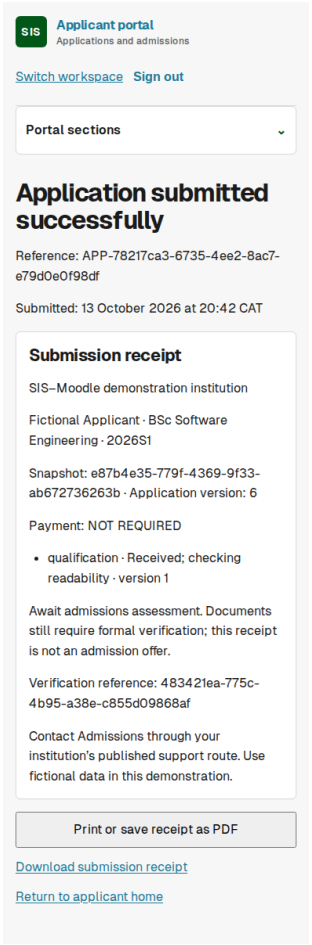

Complete capture. Enlarged page details follow the references.

# 4.2 APPLICATION DRAFT



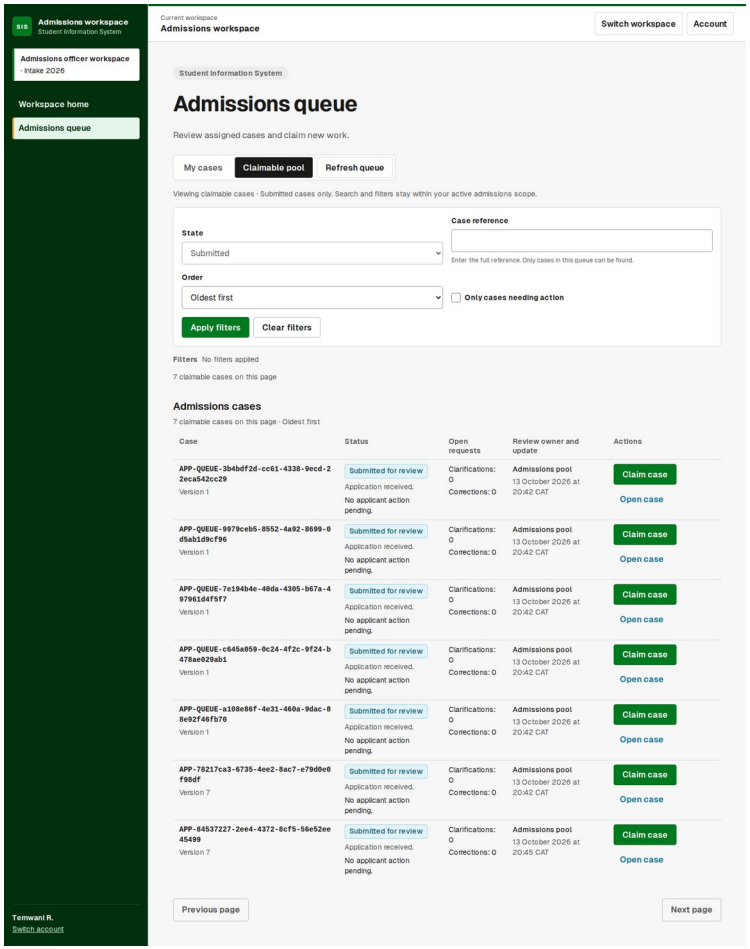

Complete capture. Enlarged page details follow the references.

# 4.3 SUBMISSION RECEIPT



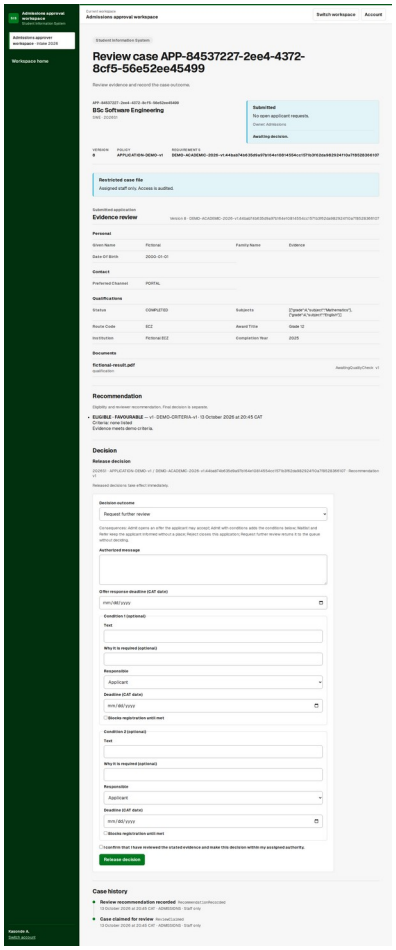

Complete capture. Enlarged page details follow the references.

# 4.4 ADMISSIONS QUEUE



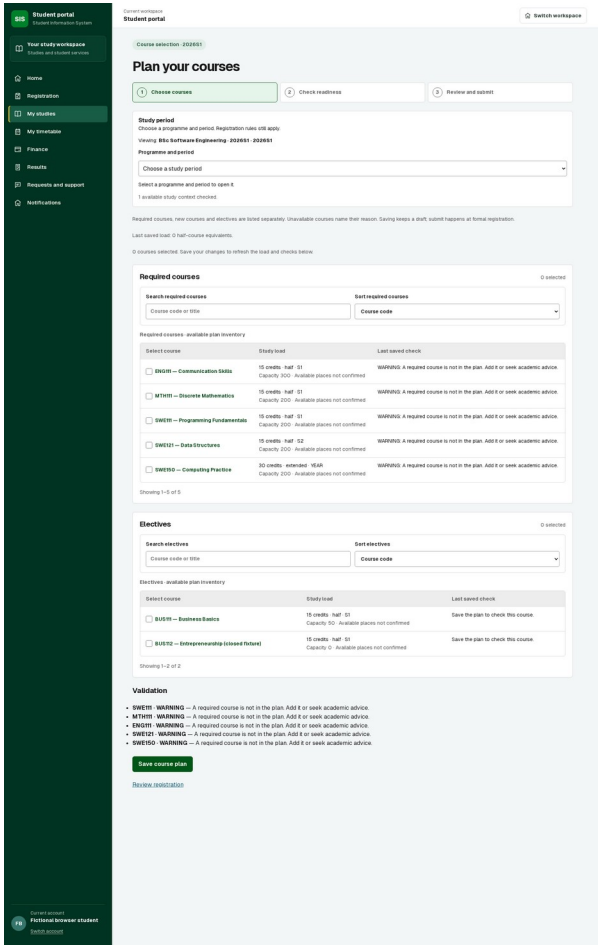

Complete capture. Enlarged page details follow the references.

# 4.5 REVIEW DECISION



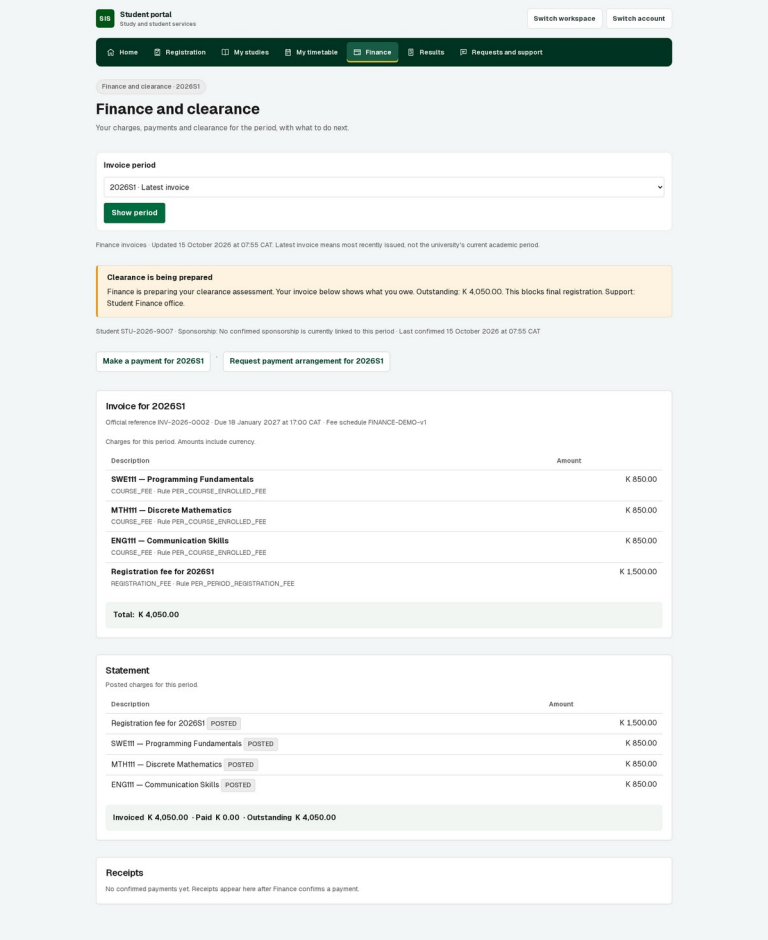

Complete capture. Enlarged page details follow the references.

# 4.6 COURSE REGISTRATION



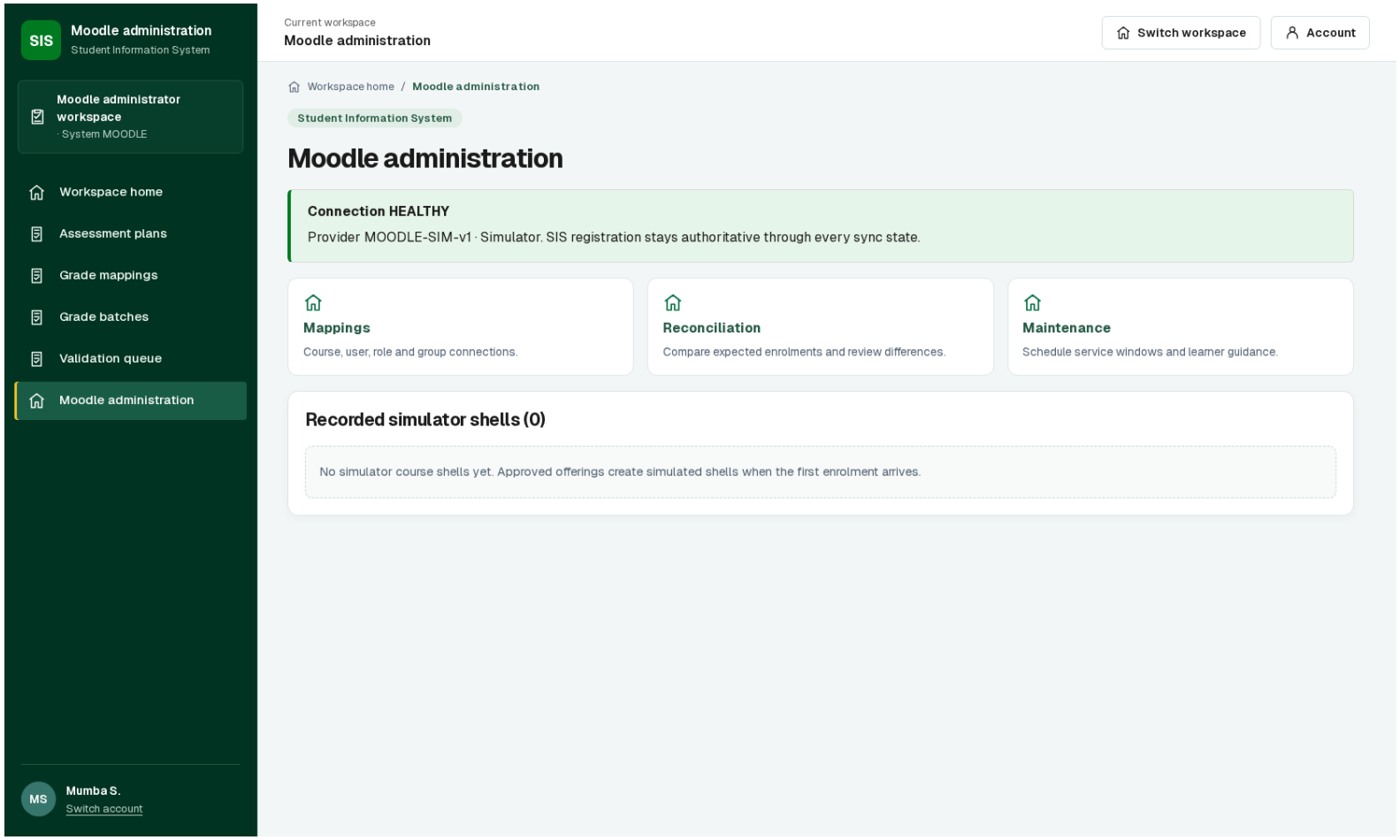

Complete capture. Enlarged page details follow the references.

# 4.7 STUDENT FINANCE



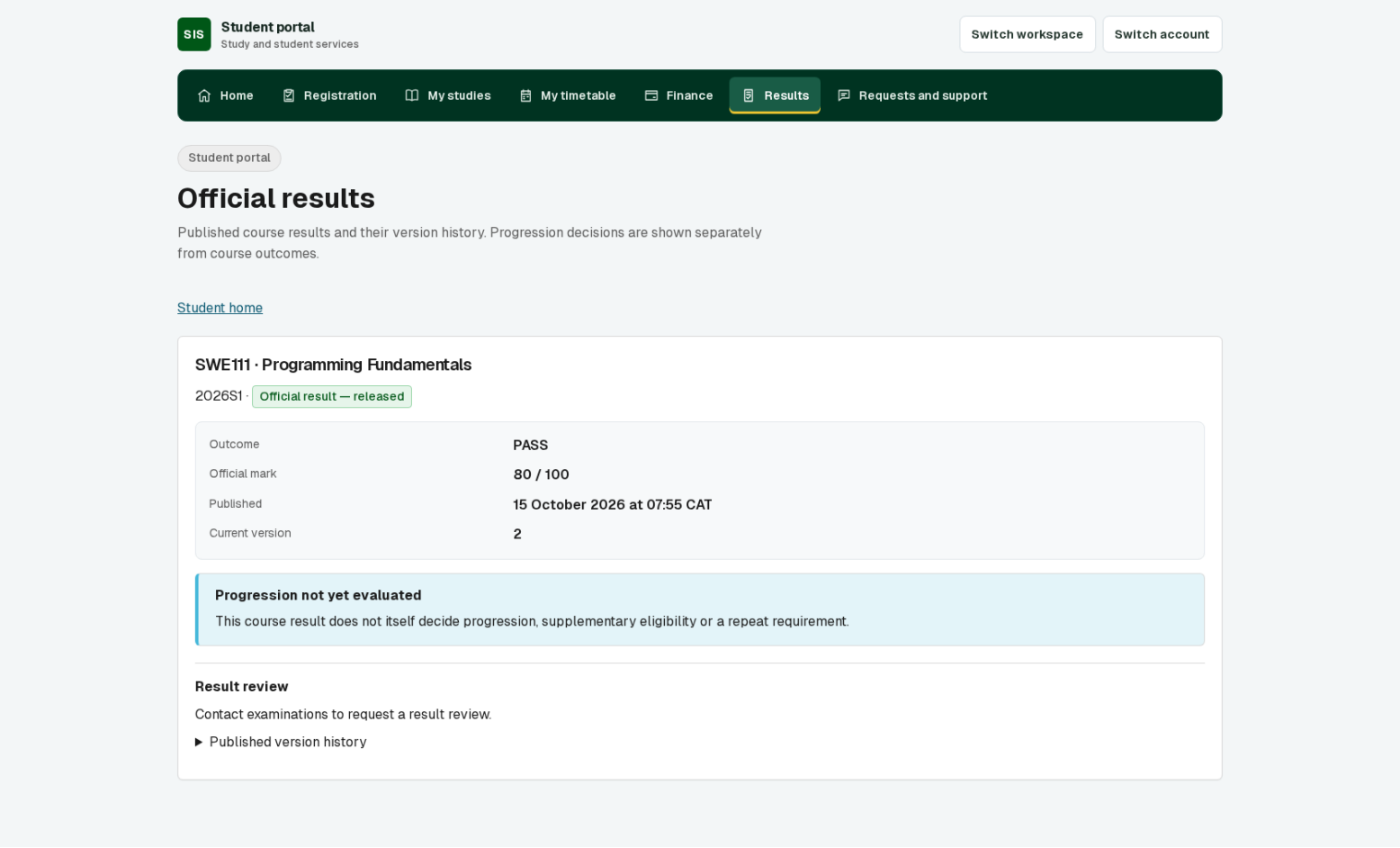

Complete capture. Enlarged page details follow the references.

# 4.8 MOODLE DASHBOARD



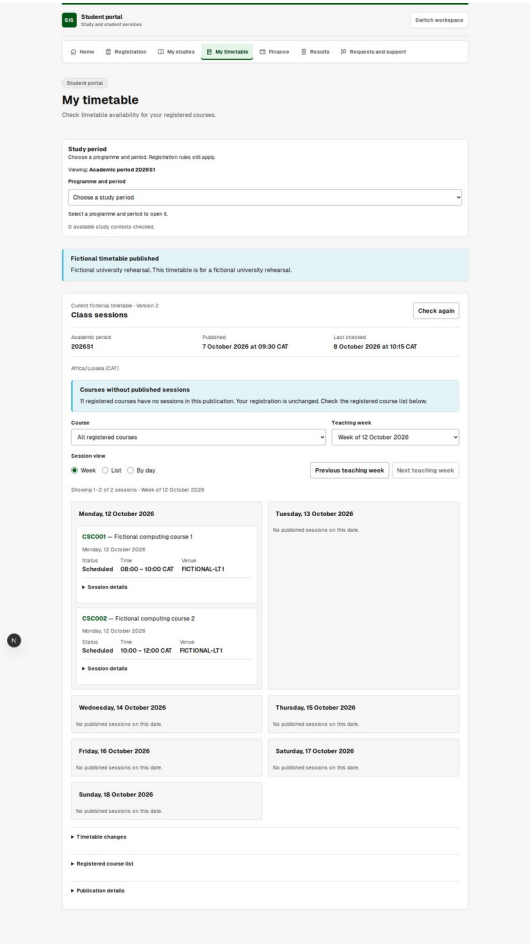

Complete capture. Enlarged page details follow the references.

# 4.9 ASSESSMENT RESULTS



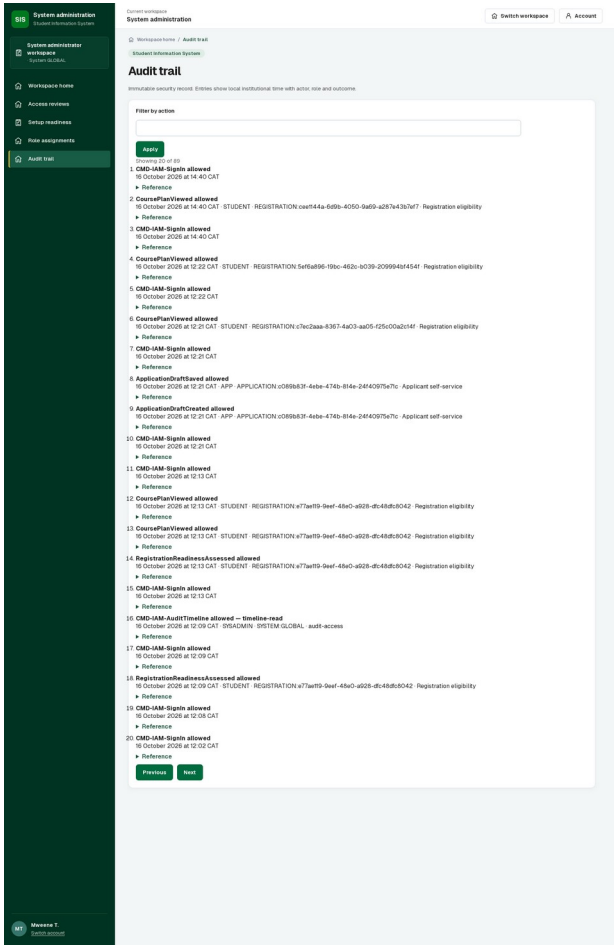

Complete capture. Enlarged page details follow the references.

# 4.10 STUDENT TIMETABLE




Complete capture. Enlarged page details follow the references.

# 4.11 AUDIT AND OPERATIONS




Complete capture. Enlarged page details follow the references.

# CONCLUSION

- We built a modular SIS with controlled registration, assessment and timetable workflows.
- We demonstrated external MoodleCloud course creation and learner access.
- Our main contribution is the separation of authoritative academic decisions from recoverable integration work.
- **The project remains a demonstrator with identified integration and AI work outstanding.**

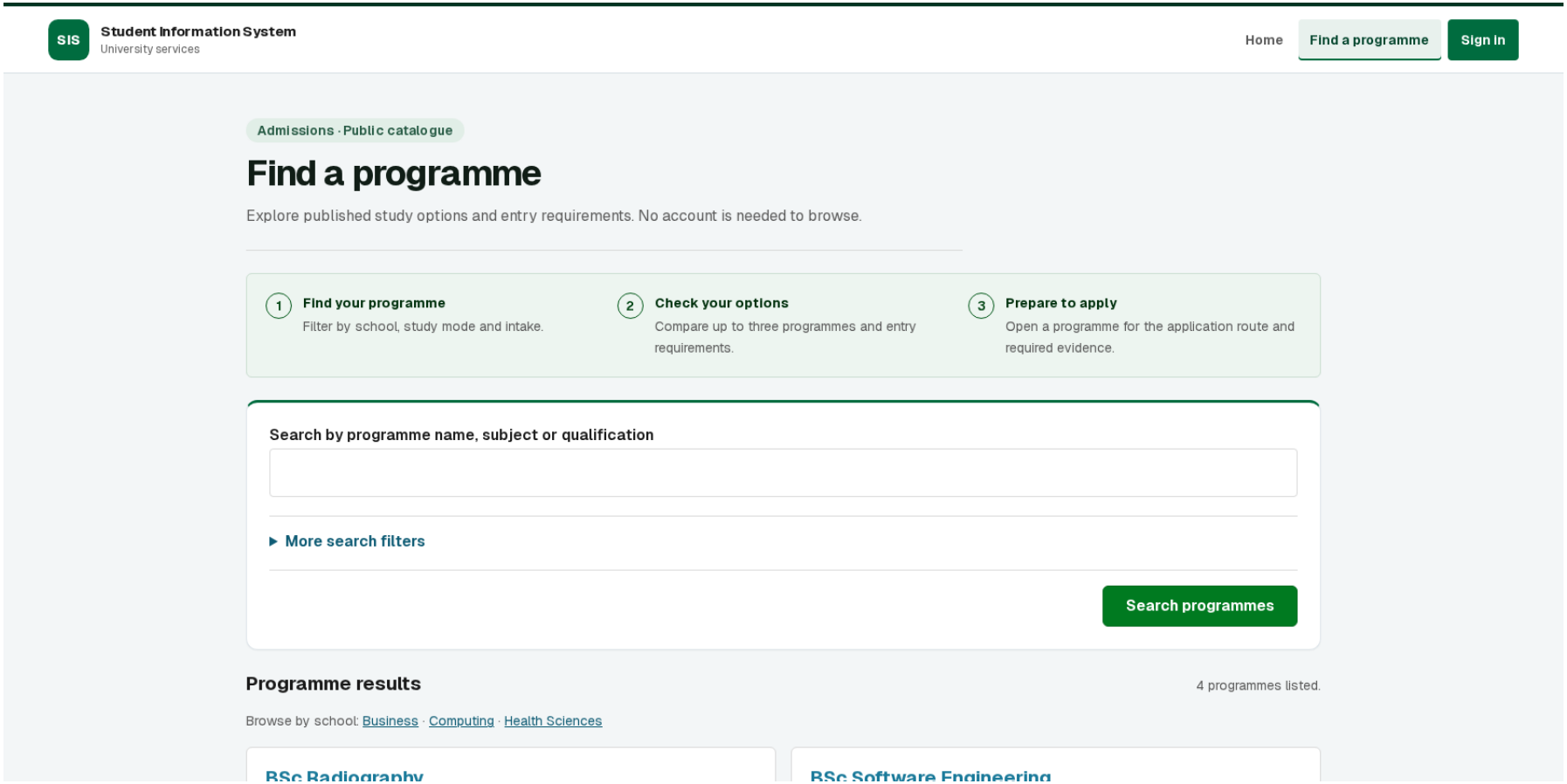
# RECOMMENDATIONS

- Complete automatic Moodle delivery and reconciliation before claiming end-to-end live synchronisation.
- Implement and evaluate LTI launch and the governed student assistant.
- Conduct supervised user acceptance, accessibility and recovery exercises.
- Confirm institutional policies before operational adoption.

# REFERENCES

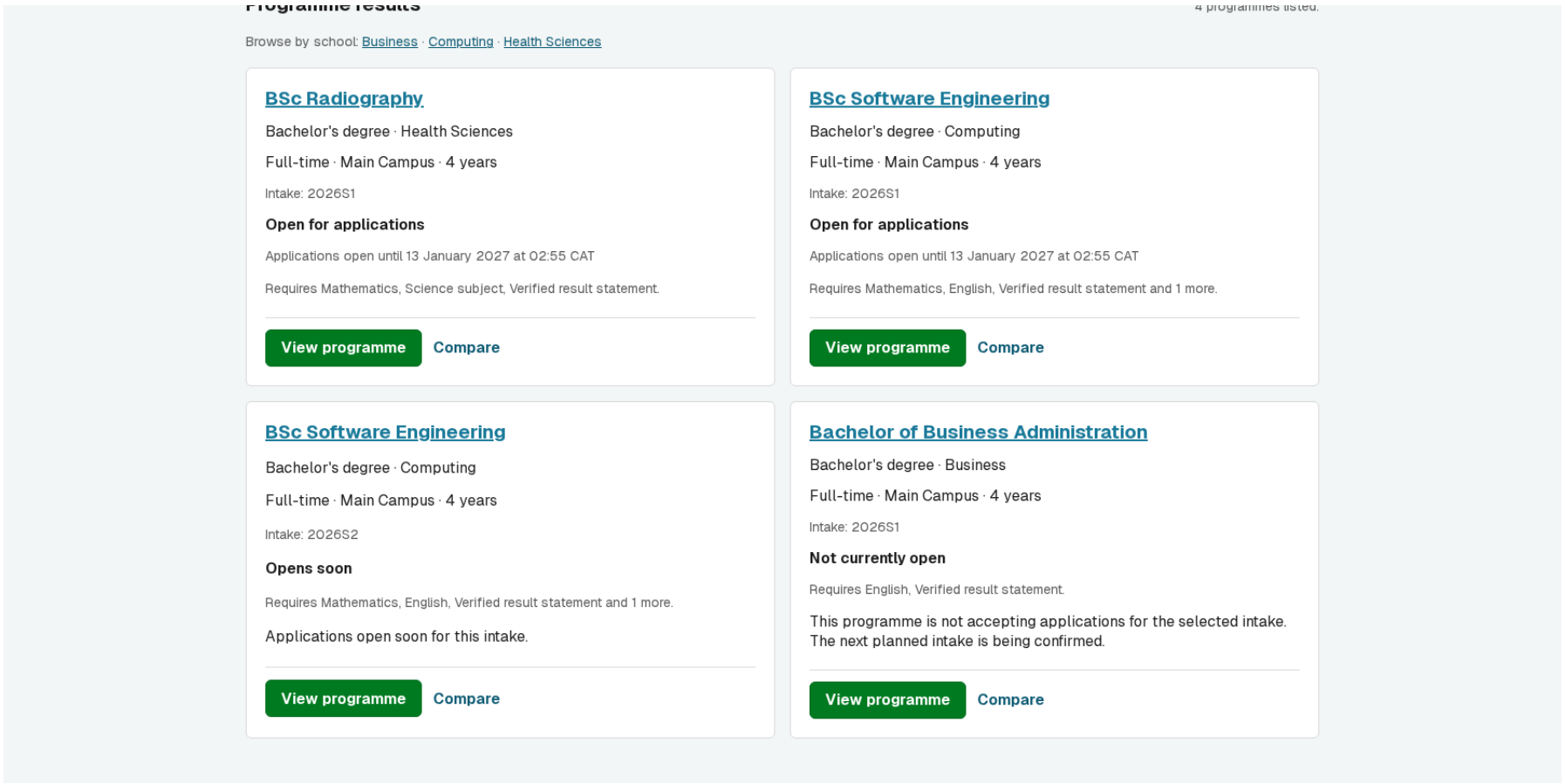

1. Moodle. Web services and External Services API documentation. MoodleDocs and Moodle Developer Resources. Accessed 16 October 2026.
2. 1EdTech Consortium. Learning Tools Interoperability Core Specification 1.3. Final Release, 2019.
3. NIST. Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1, 2023. DOI: 10.6028/NIST.AI.100-1.
4. Project evidence: our handbook, implementation records and verification logs, checkpoint 16 October 2026.

# CATALOGUE 1/2



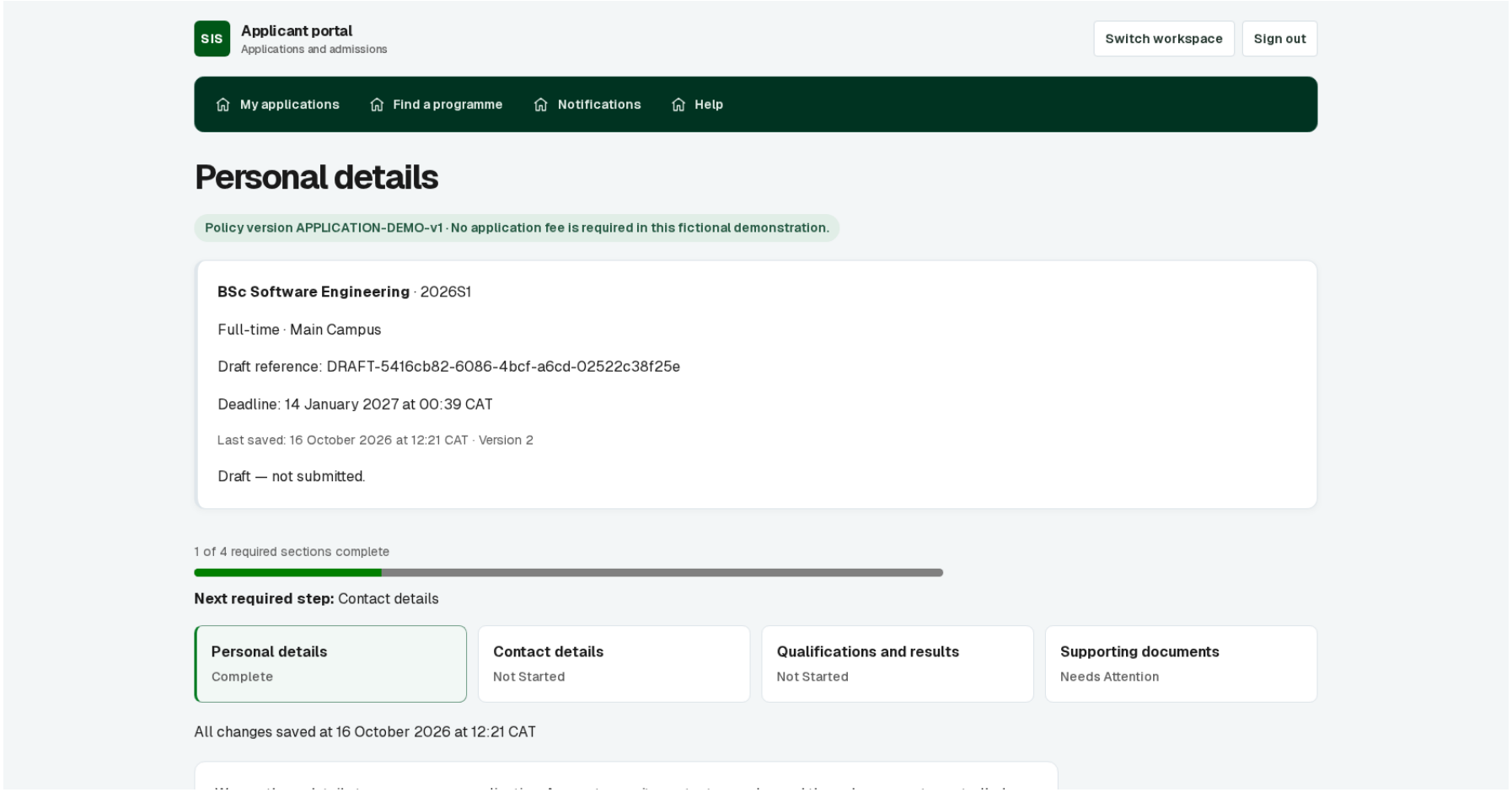

Reference appendix · full-width page continuation 1 of 2

# CATALOGUE 2/2



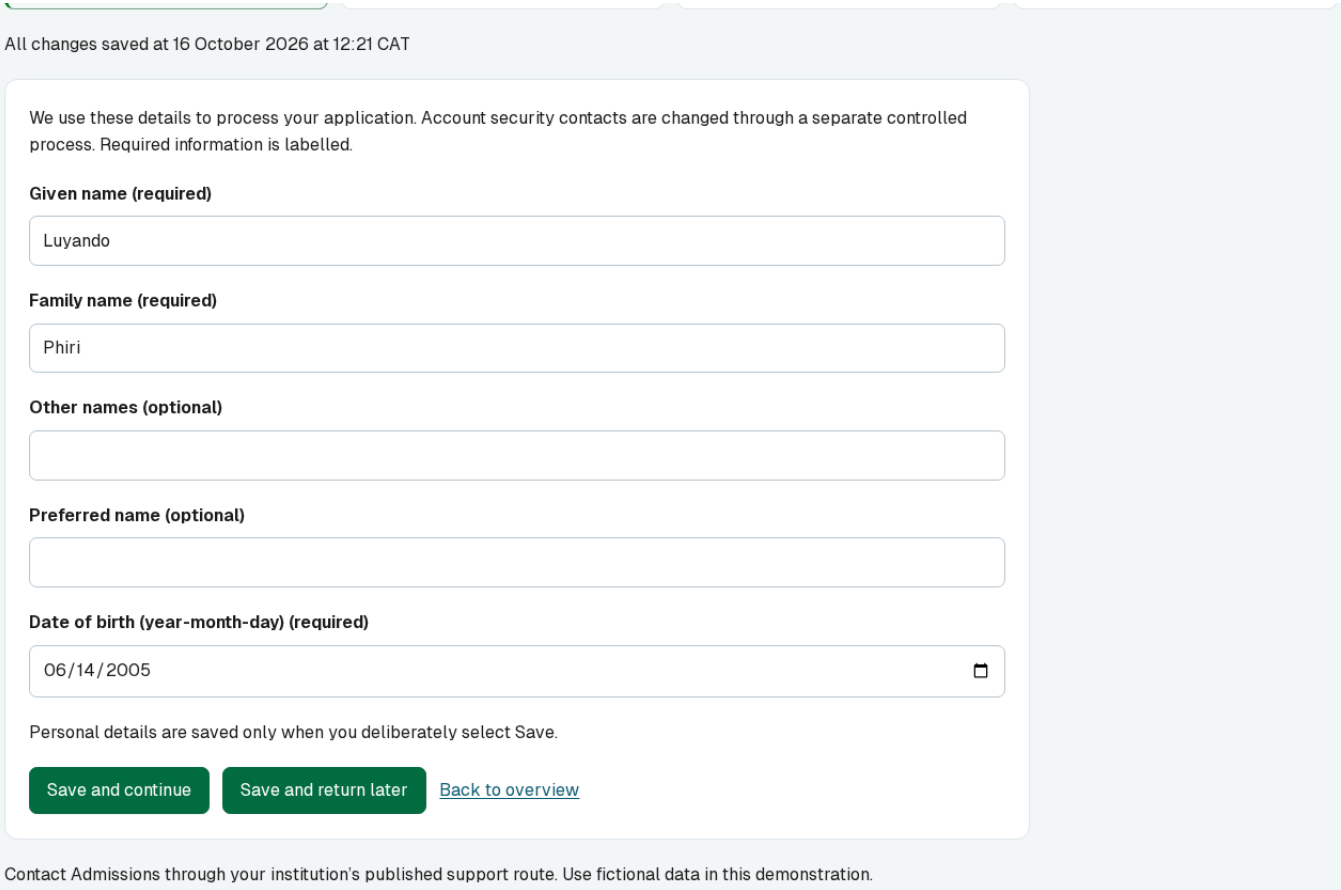

Reference appendix · full-width page continuation 2 of 2

# APPLICATION 1/2



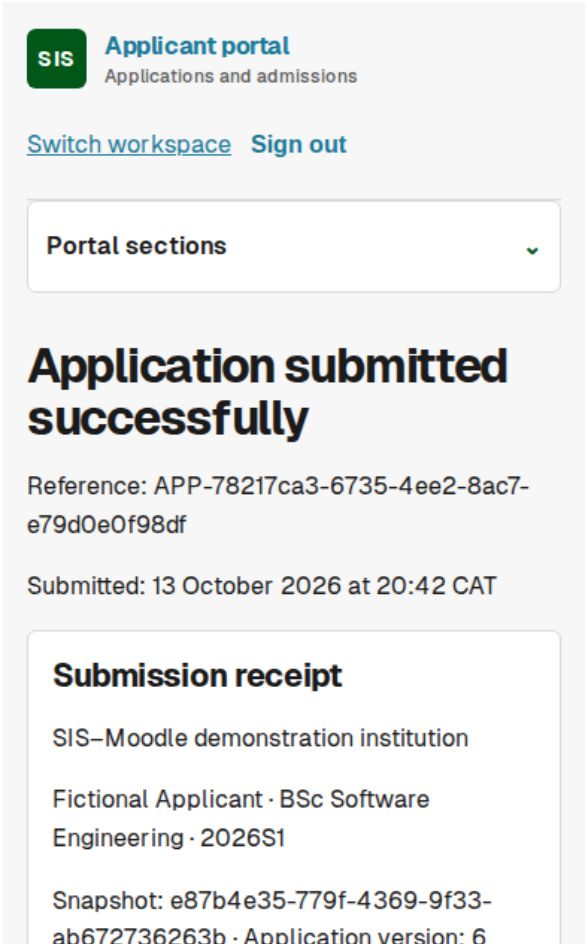

Reference appendix · full-width page continuation 1 of 2

# APPLICATION 2/2



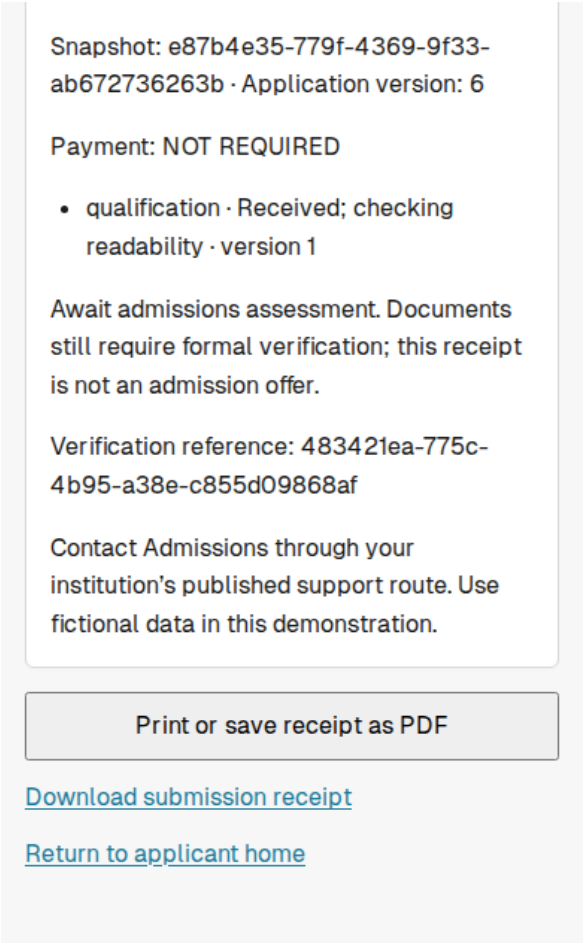

Reference appendix · full-width page continuation 2 of 2

# RECEIPT 1/2



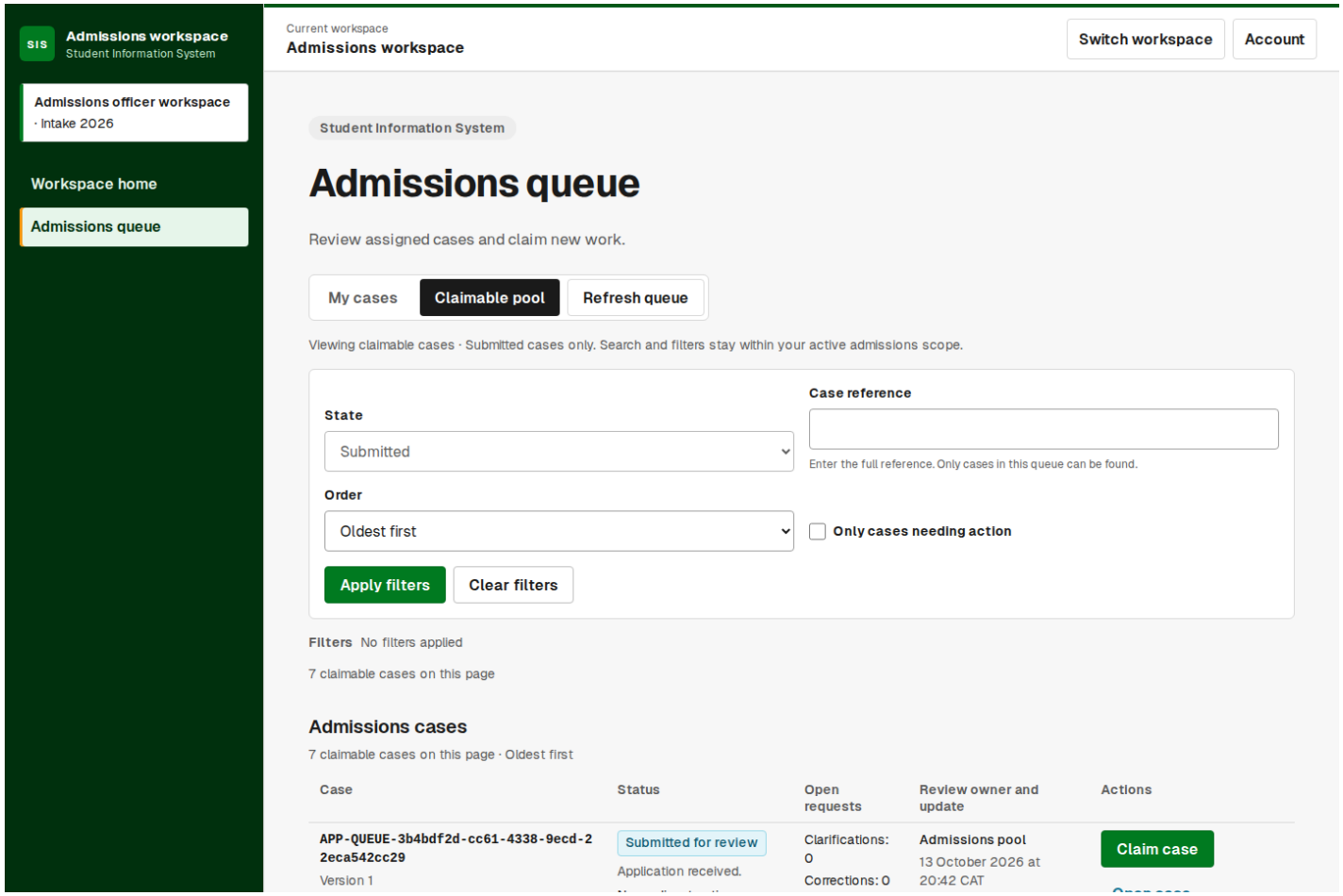

Reference appendix · full-width page continuation 1 of 2

# RECEIPT 2/2



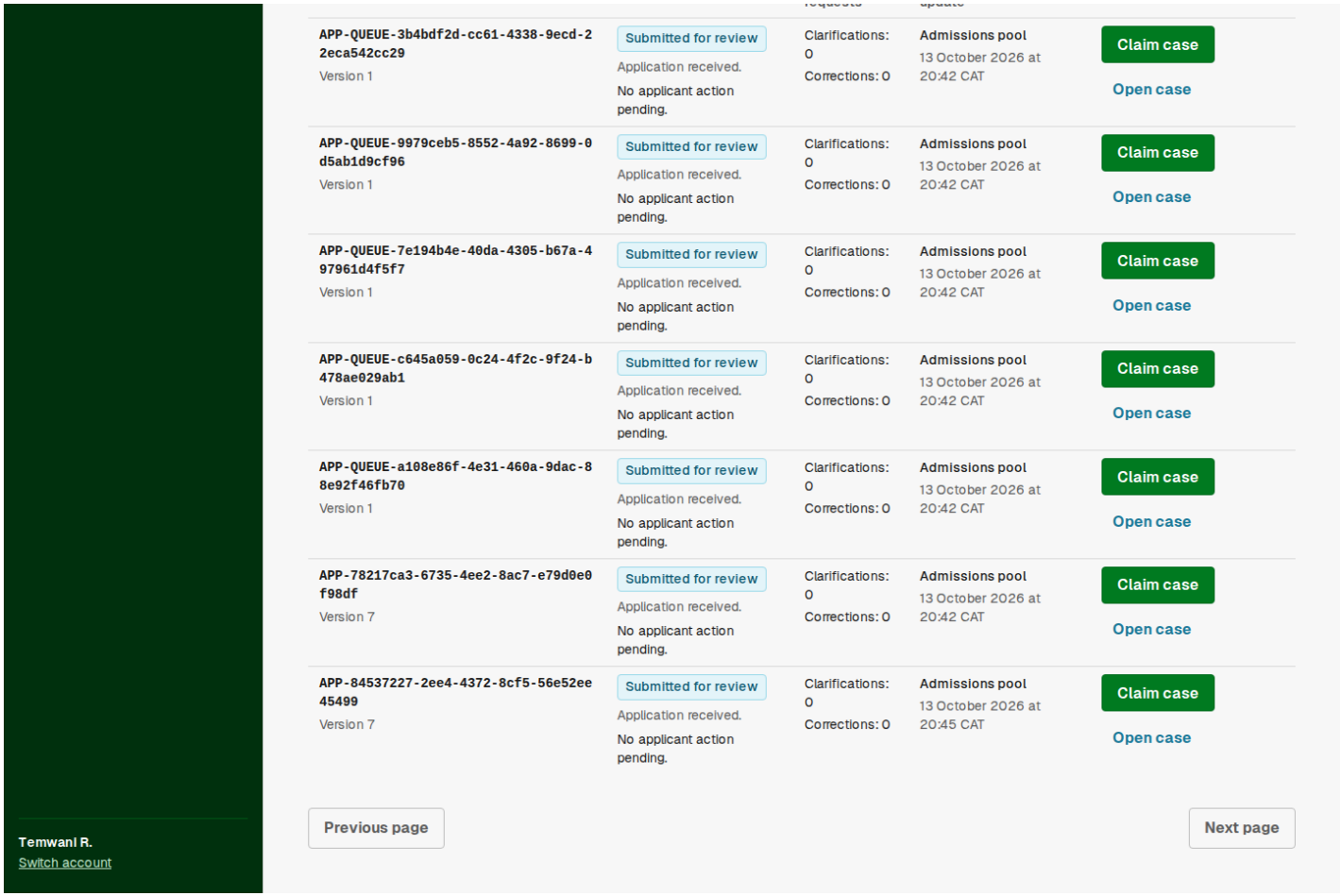

Reference appendix · full-width page continuation 2 of 2

# ADMISSIONS QUEUE 1/2



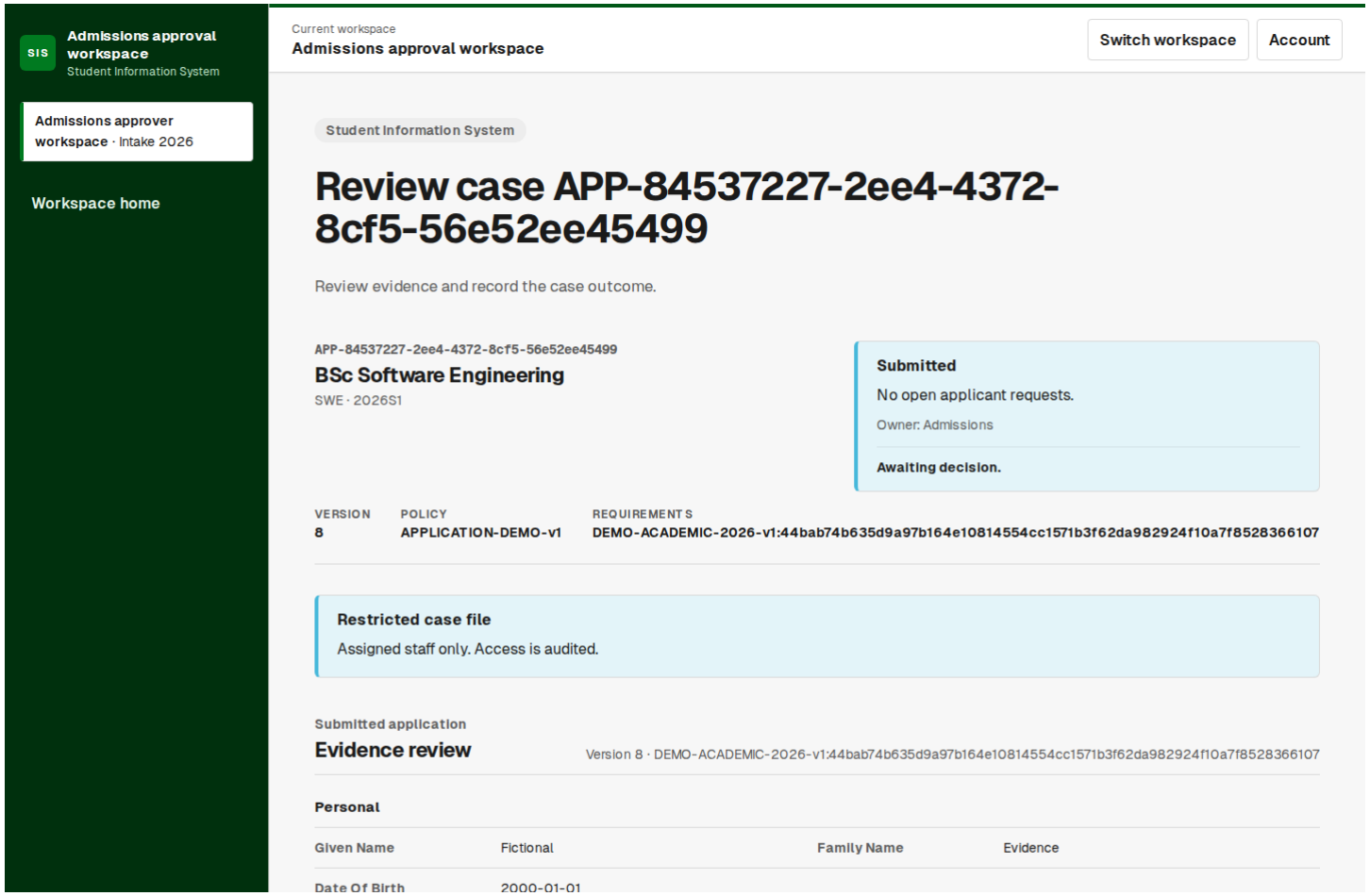

Reference appendix · full-width page continuation 1 of 2

# ADMISSIONS QUEUE 2/2



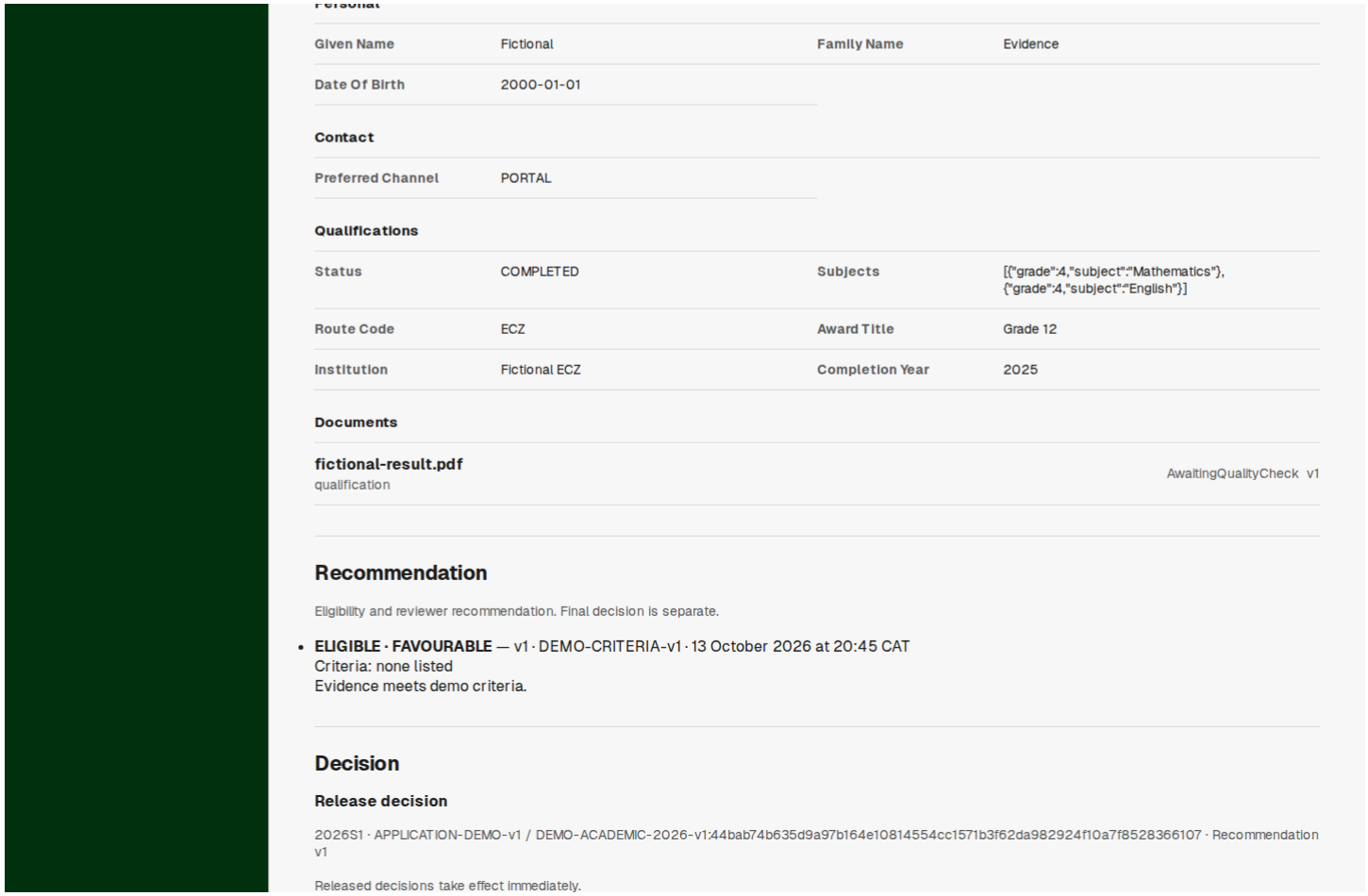

Reference appendix · full-width page continuation 2 of 2

# REVIEW DECISION 1/4




Reference appendix · full-width page continuation 1 of 4

# REVIEW DECISION 2/4



Personal

| | | | |
|---|---|---|---|
| Given Name | Fictional | Family Name | Evidence |
| Date Of Birth | 2000-01-01 | | |

**Contact**

| | |
|---|---|
| Preferred Channel | PORTAL |

**Qualifications**

| | | | |
|---|---|---|---|
| Status | COMPLETED | Subjects | [{"grade":4,"subject":"Mathematics"}, {"grade":4,"subject":"English"}] |
| Route Code | ECZ | Award Title | Grade 12 |
| Institution | Fictional ECZ | Completion Year | 2025 |

**Documents**

**fictional-result.pdf**
qualification
AwaitingQualityCheck v1

## Recommendation

Eligibility and reviewer recommendation. Final decision is separate.

- **ELIGIBLE · FAVOURABLE** — v1 · DEMO-CRITERIA-v1 · 13 October 2026 at 20:45 CAT
  Criteria: none listed
  Evidence meets demo criteria.

## Decision

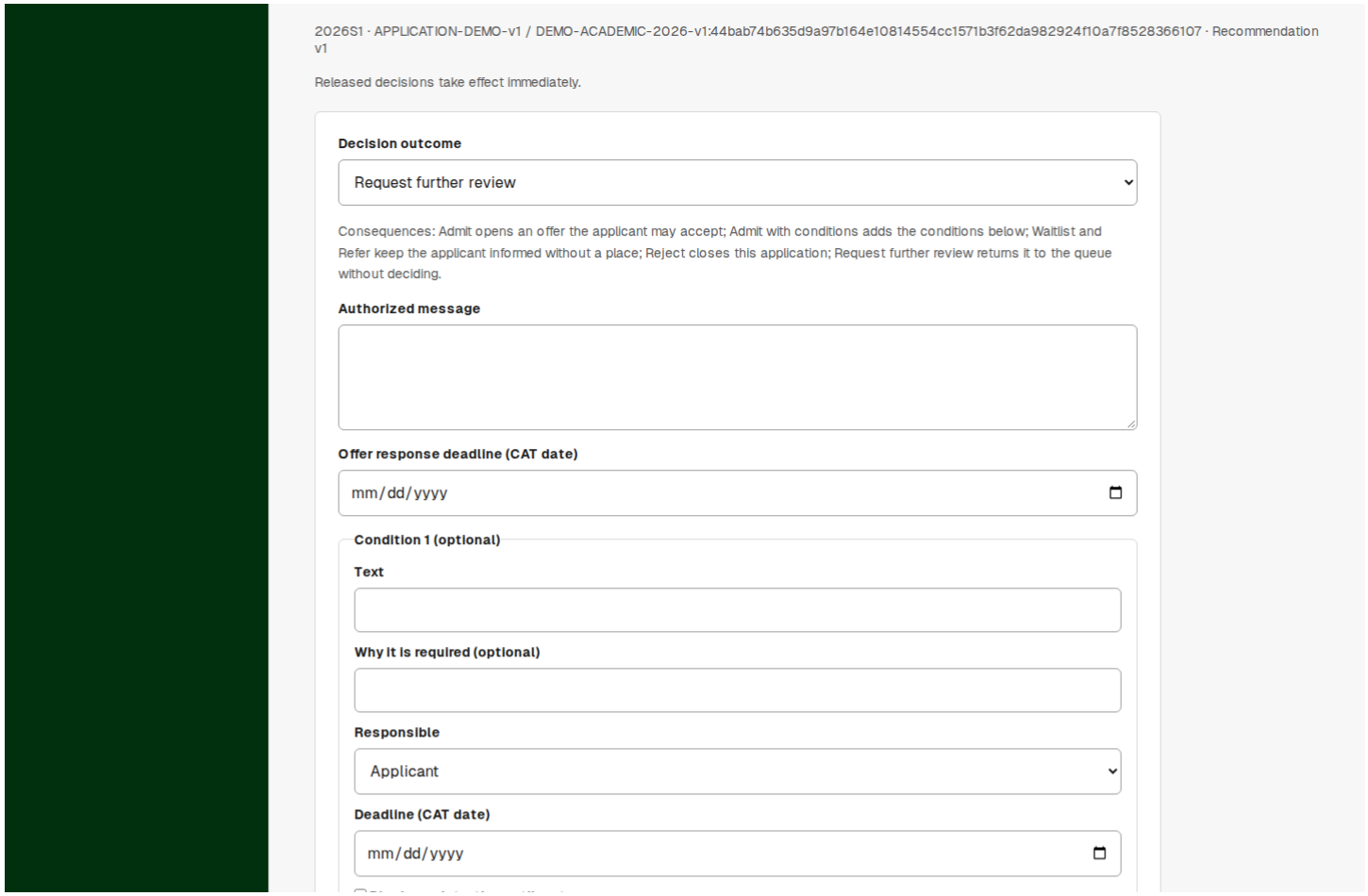
**Release decision**

2026S1 · APPLICATION-DEMO-v1 / DEMO-ACADEMIC-2026-v1:44bab74b635d9a97b164e10814554cc1571b3f62da982924f10a7f8528366107 · Recommendation v1

Released decisions take effect immediately.

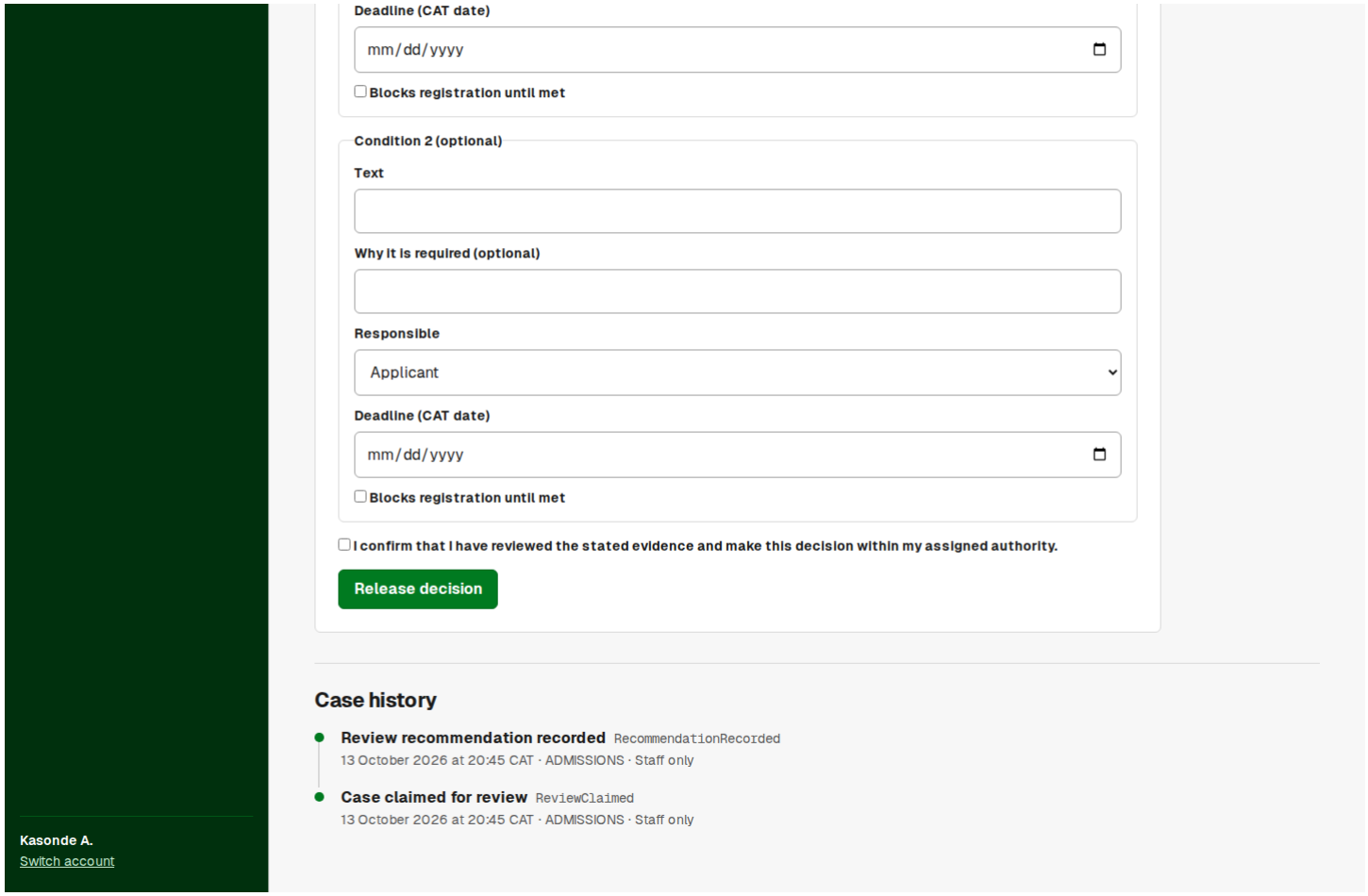
Reference appendix · full-width page continuation 2 of 4

# REVIEW DECISION 3/4



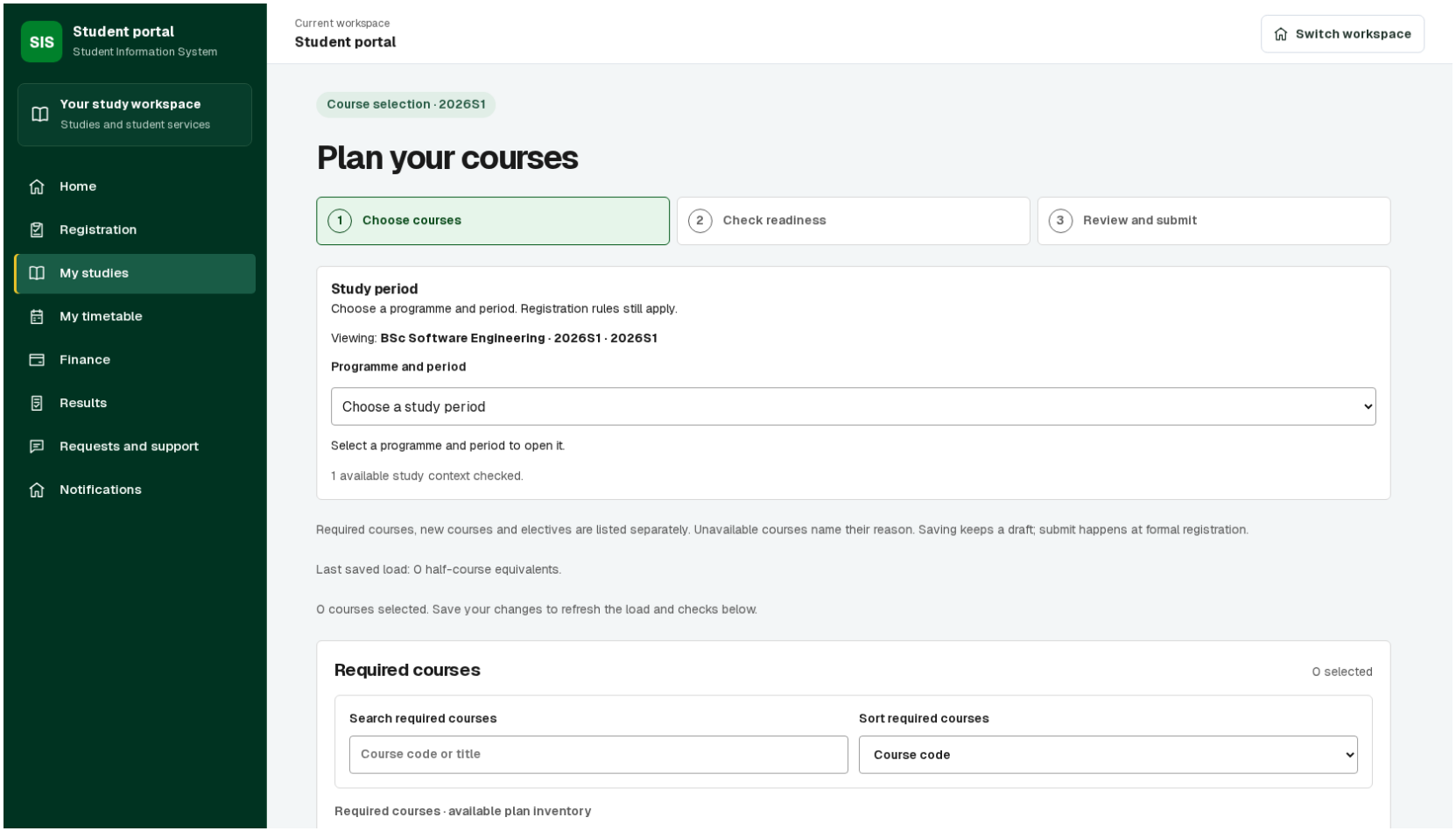

Reference appendix · full-width page continuation 3 of 4

# REVIEW DECISION 4/4



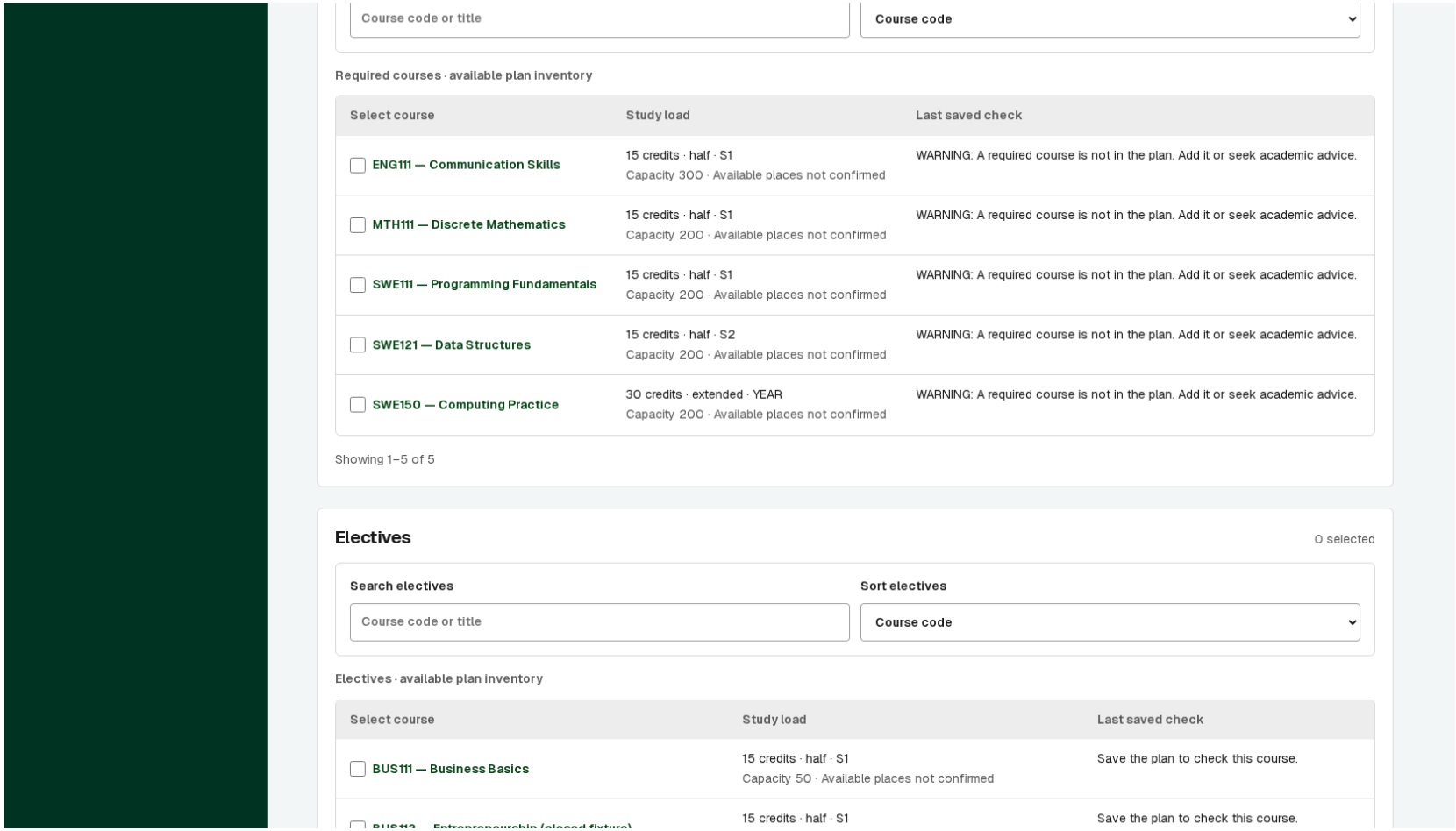

Reference appendix · full-width page continuation 4 of 4

# COURSE SELECTION 1/3



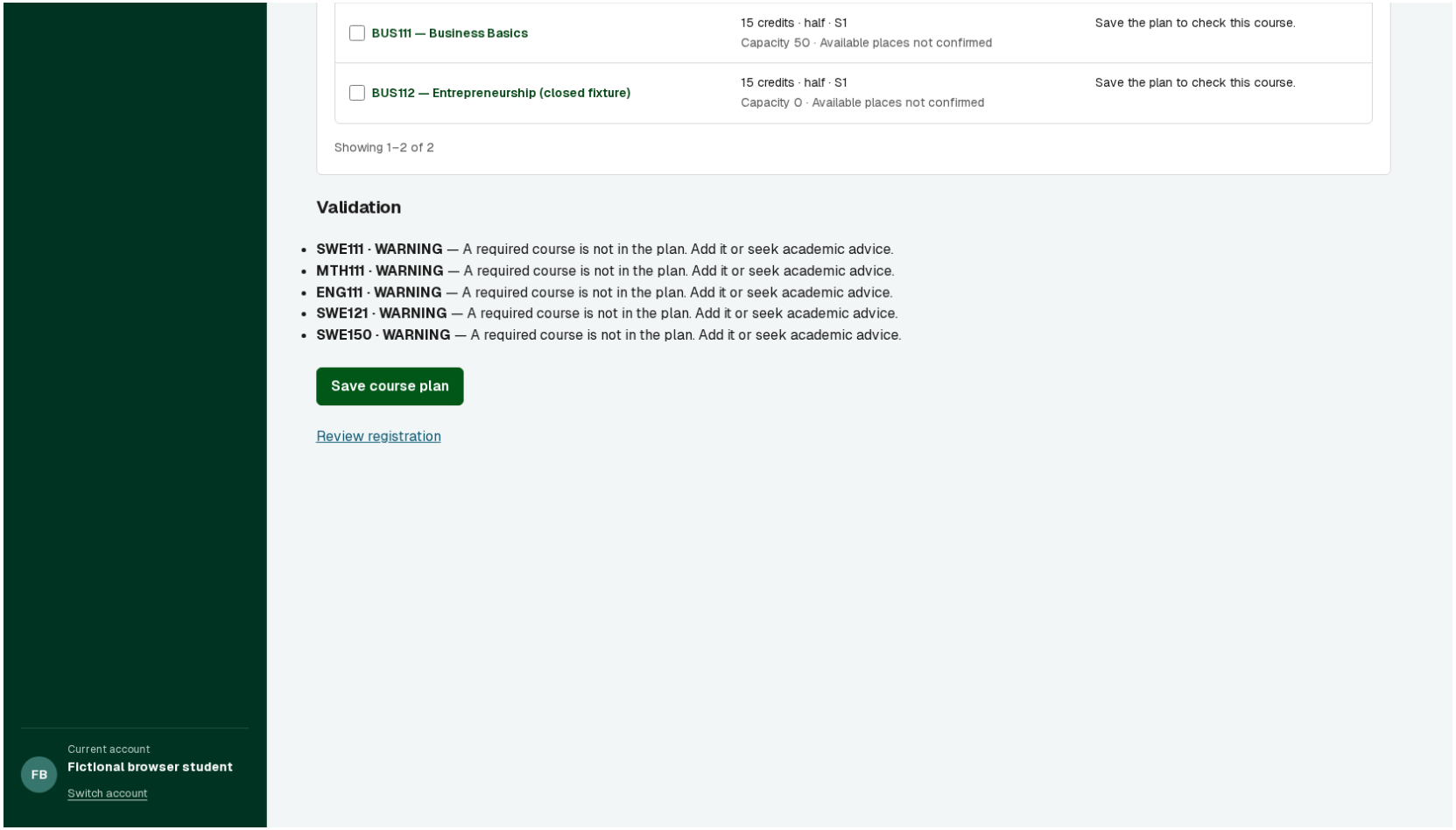

Reference appendix · full-width page continuation 1 of 3

# COURSE SELECTION 2/3



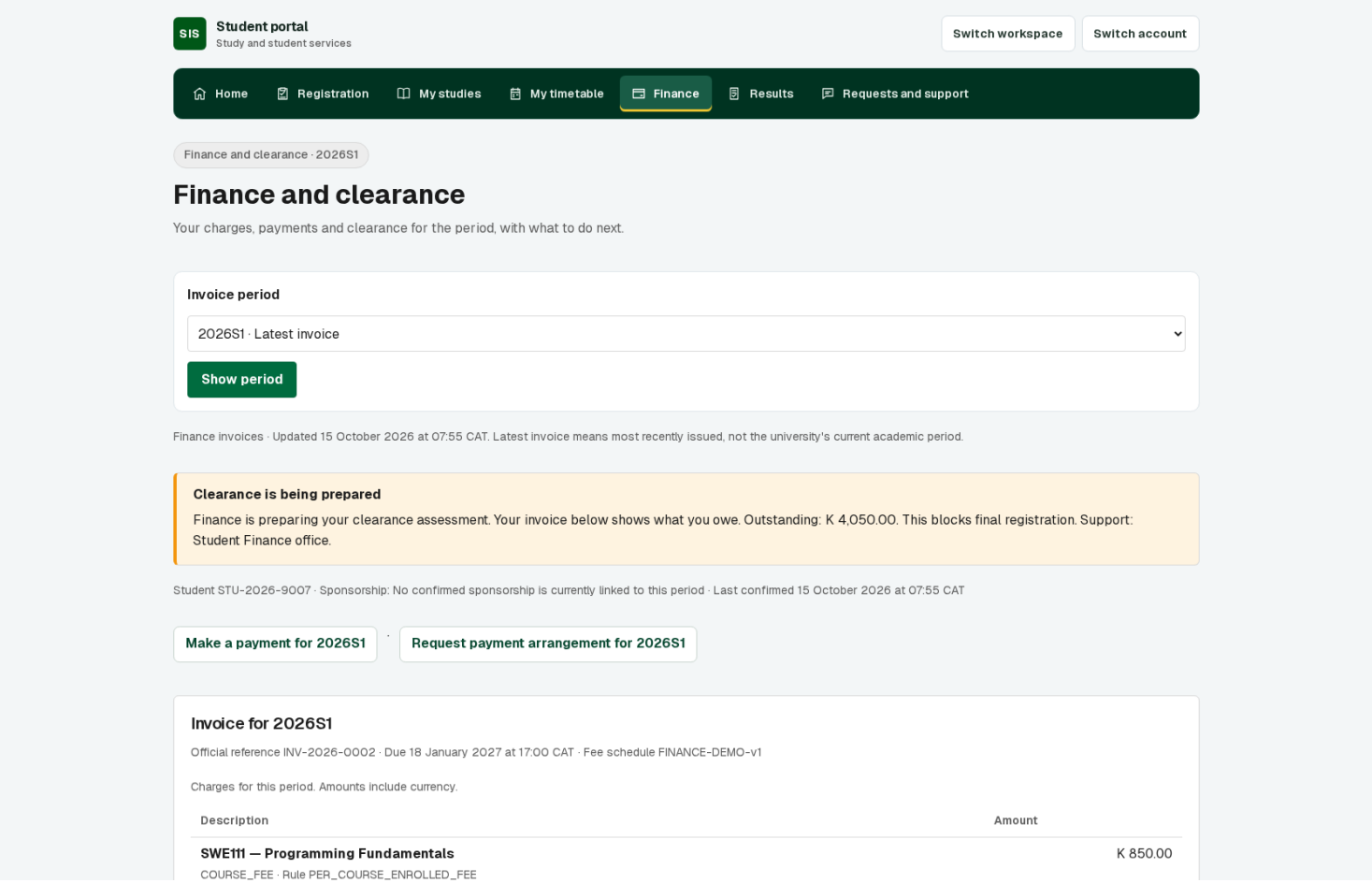

Reference appendix · full-width page continuation 2 of 3

# COURSE SELECTION 3/3



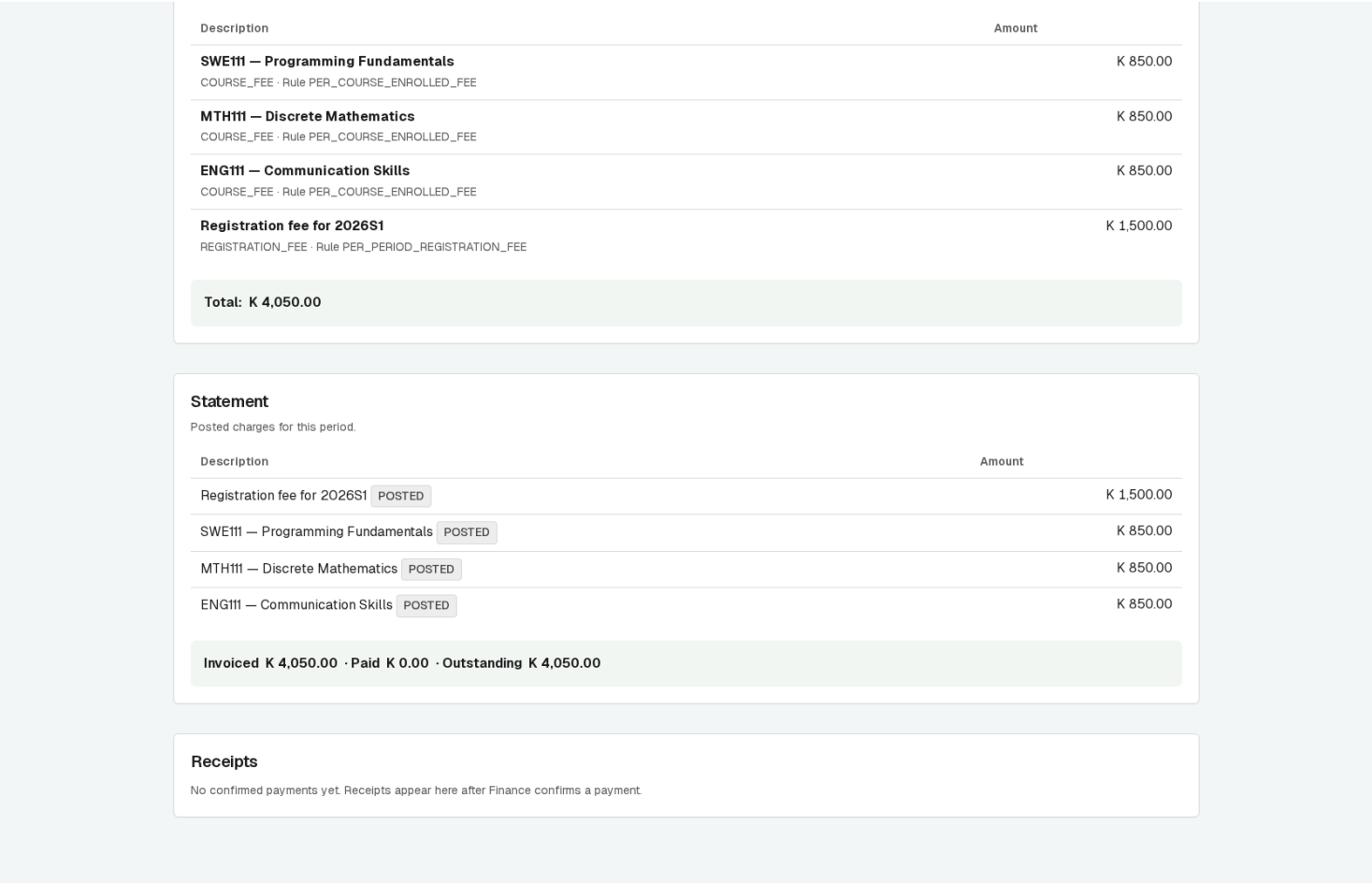

Reference appendix · full-width page continuation 3 of 3

# FINANCE 1/2




Reference appendix · full-width page continuation 1 of 2

# FINANCE 2/2



| Description | Amount |
|---|---|
| **SWE111 — Programming Fundamentals**<br>COURSE_FEE · Rule PER_COURSE_ENROLLED_FEE | K 850.00 |
| **MTH111 — Discrete Mathematics**<br>COURSE_FEE · Rule PER_COURSE_ENROLLED_FEE | K 850.00 |
| **ENG111 — Communication Skills**<br>COURSE_FEE · Rule PER_COURSE_ENROLLED_FEE | K 850.00 |
| **Registration fee for 2026S1**<br>REGISTRATION_FEE · Rule PER_PERIOD_REGISTRATION_FEE | K 1,500.00 |

**Total: K 4,050.00**

**Statement**

Posted charges for this period.

| Description | Amount |
|---|---|
| Registration fee for 2026S1 POSTED | K 1,500.00 |
| SWE111 — Programming Fundamentals POSTED | K 850.00 |
| MTH111 — Discrete Mathematics POSTED | K 850.00 |
| ENG111 — Communication Skills POSTED | K 850.00 |

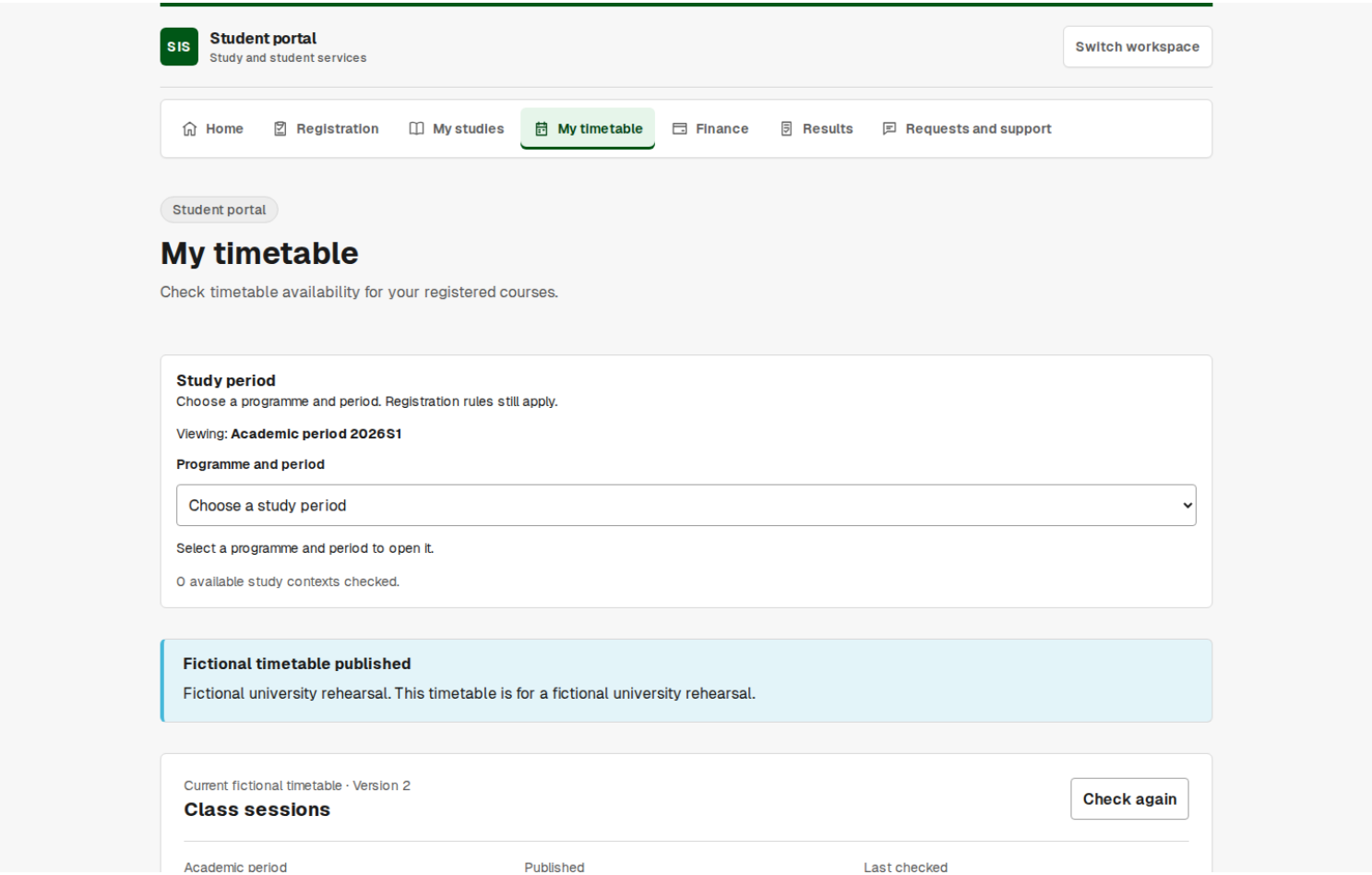
**Invoiced K 4,050.00 · Paid K 0.00 · Outstanding K 4,050.00**

**Receipts**

No confirmed payments yet. Receipts appear here after Finance confirms a payment.

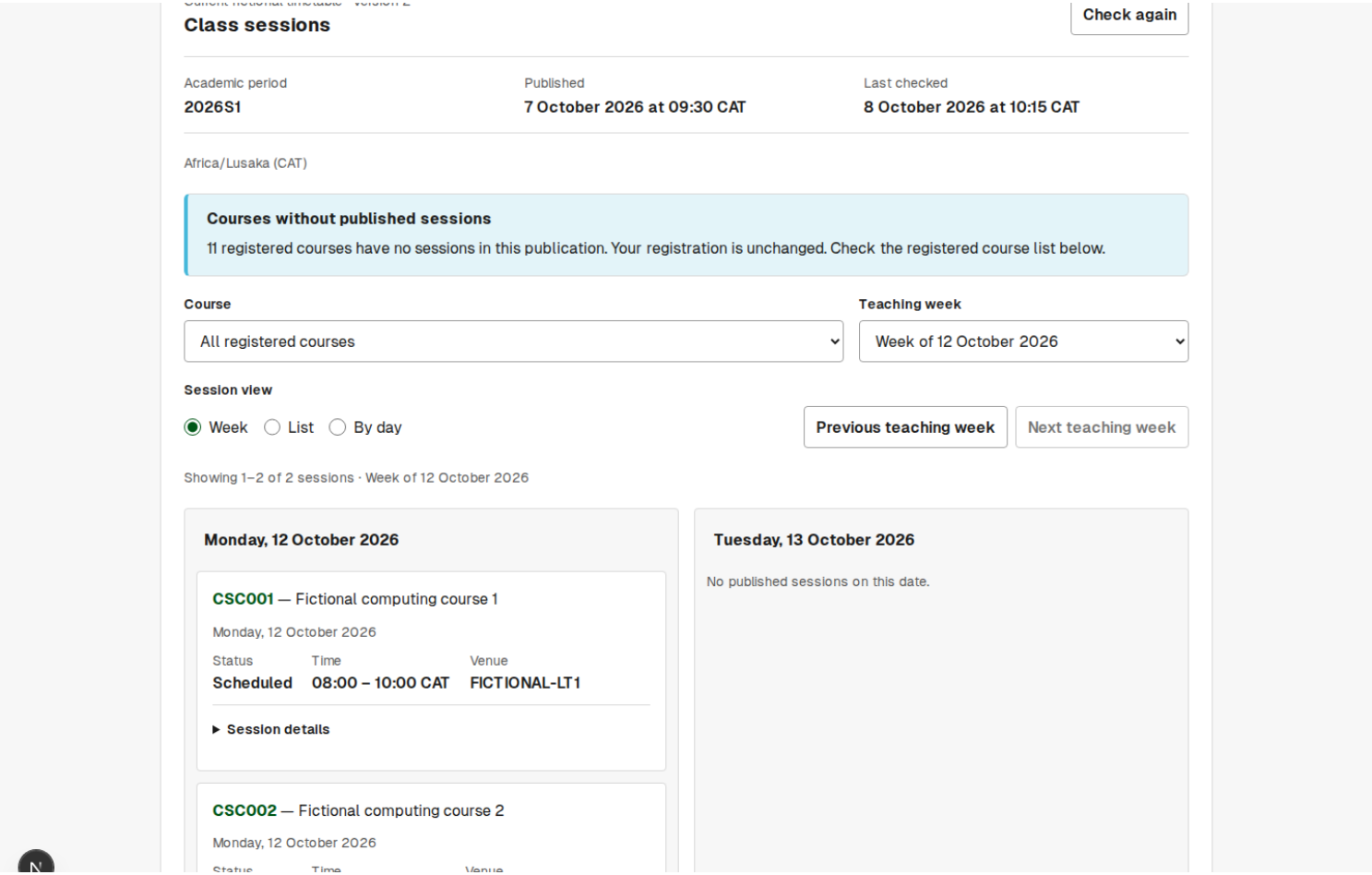
Reference appendix · full-width page continuation 2 of 2

# TIMETABLE 1/3




Reference appendix · full-width page continuation 1 of 3

# TIMETABLE 2/3



## Class sessions

Check again

| Academic period | Published | Last checked |
|---|---|---|
| 2026S1 | 7 October 2026 at 09:30 CAT | 8 October 2026 at 10:15 CAT |

Africa/Lusaka (CAT)

**Courses without published sessions**

11 registered courses have no sessions in this publication. Your registration is unchanged. Check the registered course list below.

Course: All registered courses

Teaching week: Week of 12 October 2026

Session view: (•) Week ( ) List ( ) By day

Previous teaching week | Next teaching week

Showing 1–2 of 2 sessions · Week of 12 October 2026

**Monday, 12 October 2026**

**CSC001** — Fictional computing course 1

Monday, 12 October 2026

| Status | Time | Venue |
|---|---|---|
| Scheduled | 08:00 – 10:00 CAT | FICTIONAL-LT1 |

▸ Session details

**CSC002** — Fictional computing course 2

Monday, 12 October 2026

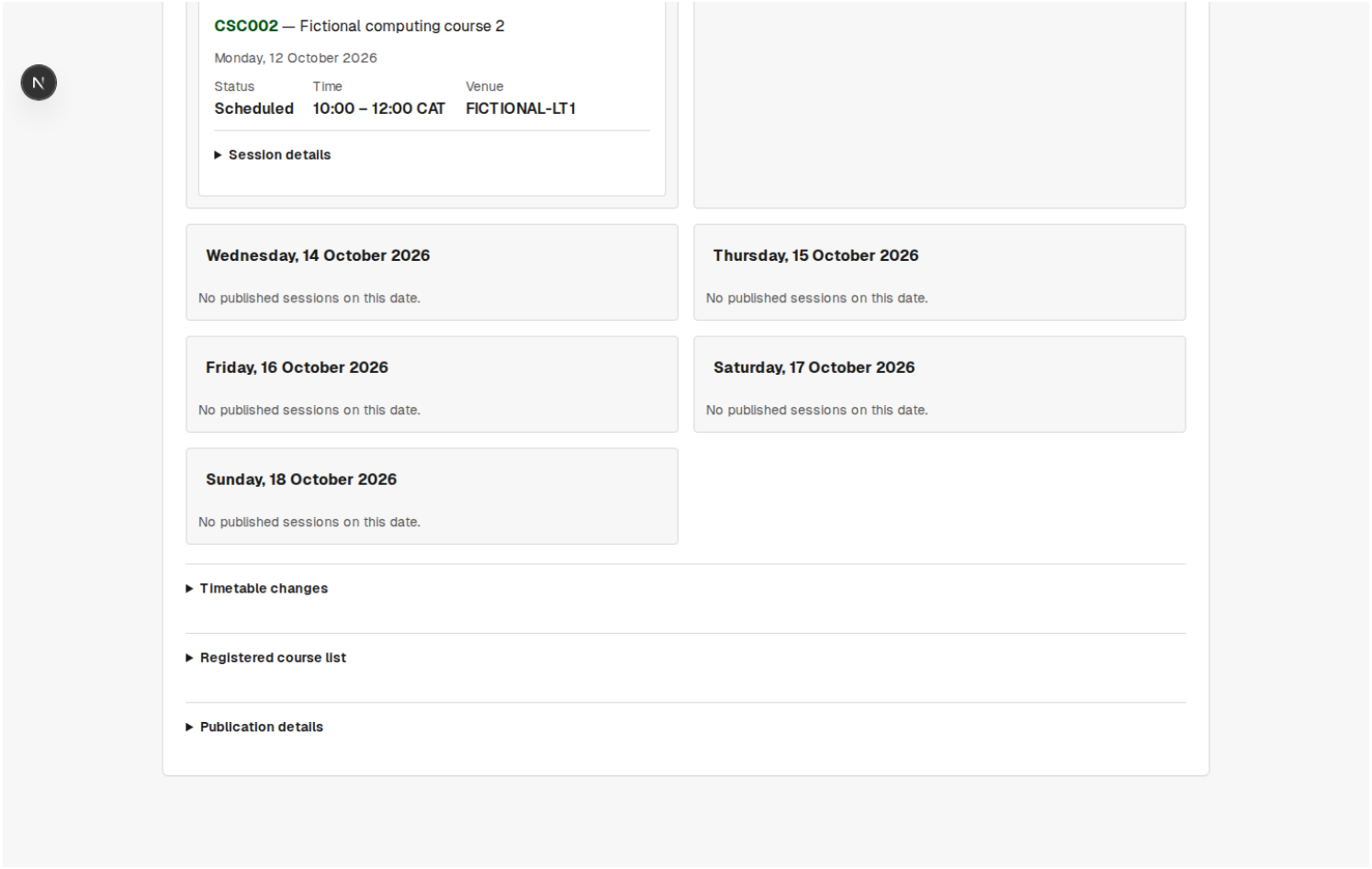
Status Time Venue

**Tuesday, 13 October 2026**

No published sessions on this date.

Reference appendix · full-width page continuation 2 of 3

# TIMETABLE 3/3



**CSC002** — Fictional computing course 2

Monday, 12 October 2026

| Status | Time | Venue |
|---|---|---|
| Scheduled | 10:00 – 12:00 CAT | FICTIONAL-LT1 |

▸ Session details

**Wednesday, 14 October 2026**

No published sessions on this date.

**Thursday, 15 October 2026**

No published sessions on this date.

**Friday, 16 October 2026**

No published sessions on this date.

**Saturday, 17 October 2026**

No published sessions on this date.

**Sunday, 18 October 2026**

No published sessions on this date.

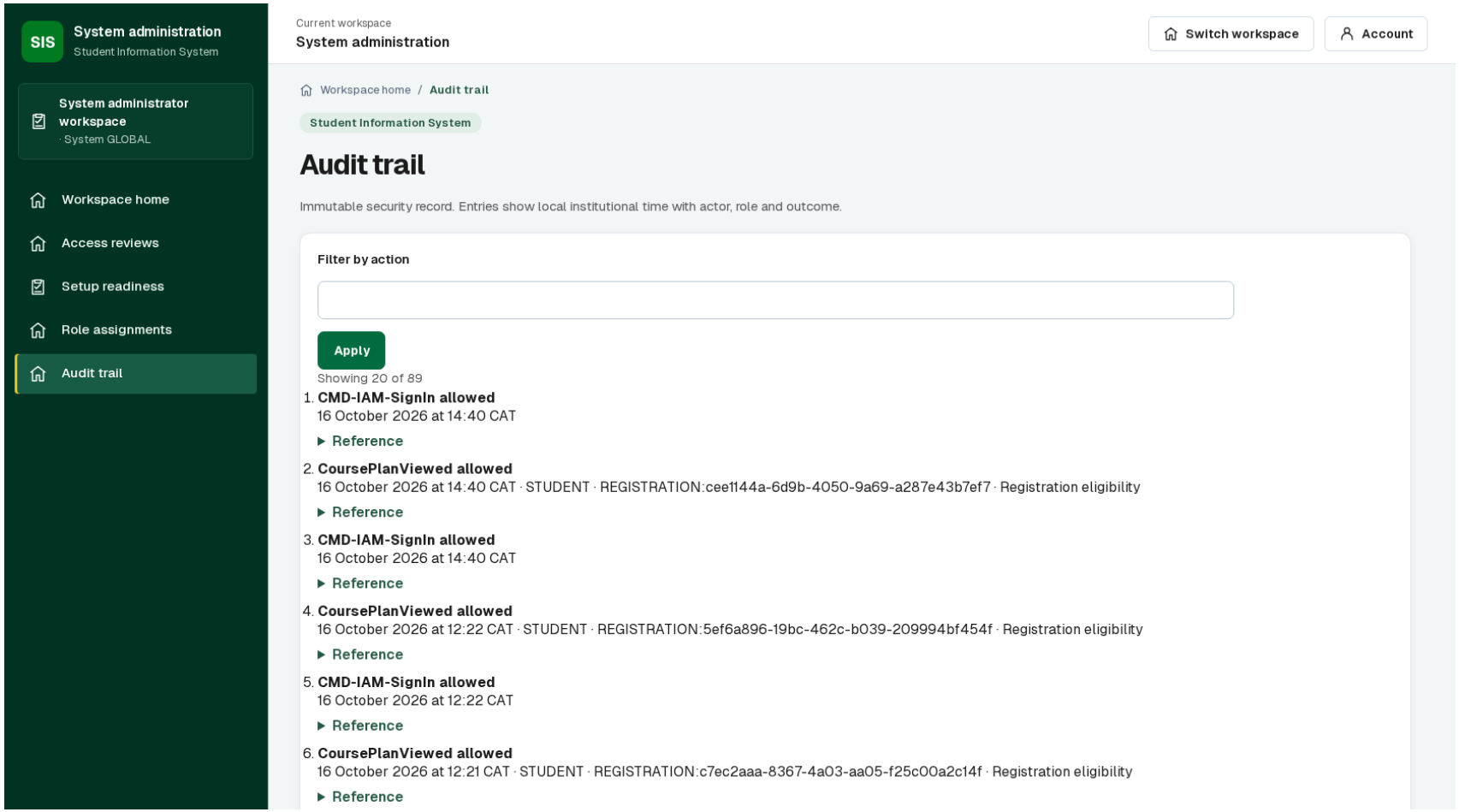
▸ Timetable changes

▸ Registered course list

▸ Publication details

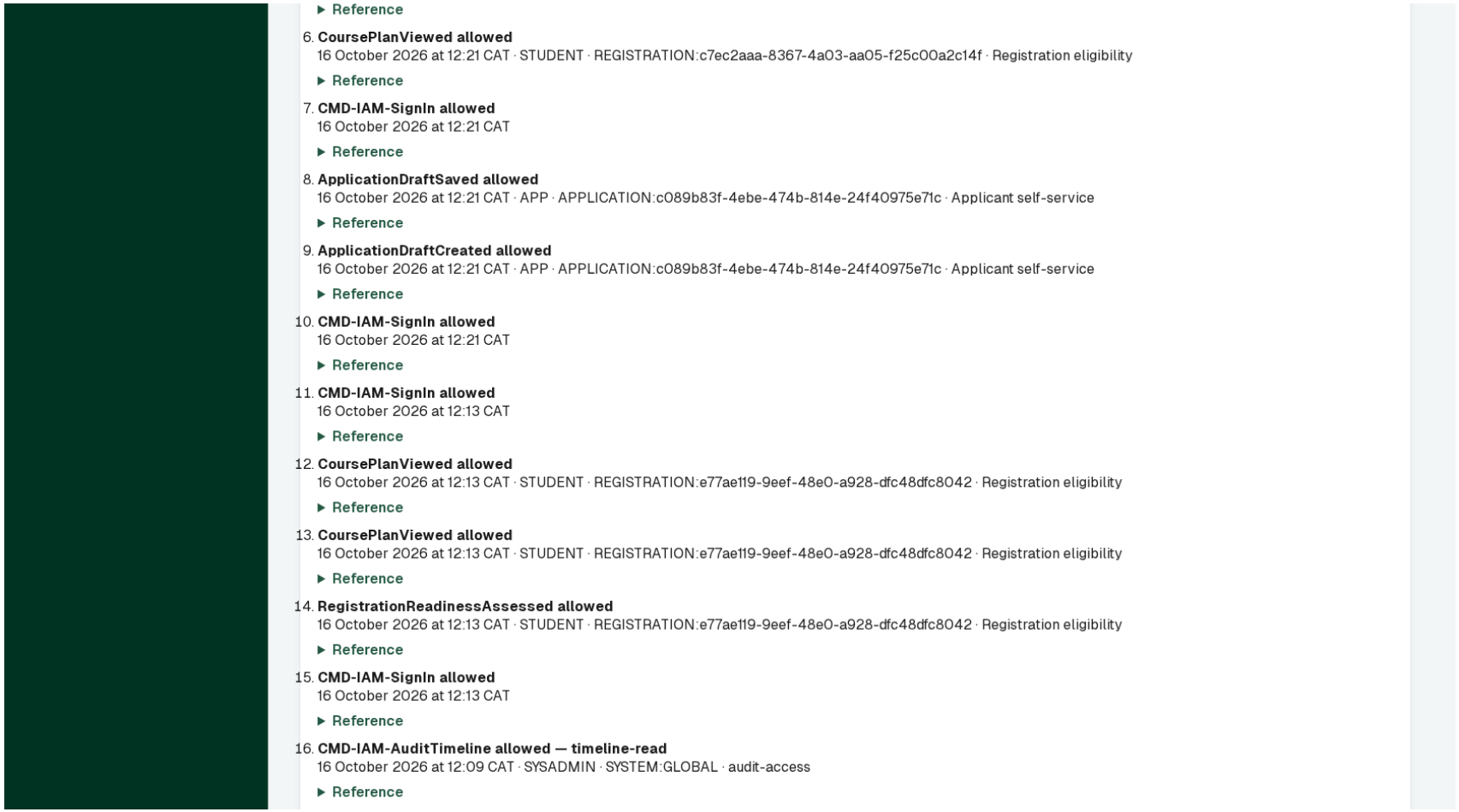
Reference appendix · full-width page continuation 3 of 3

# AUDIT TRAIL 1/3




Reference appendix · full-width page continuation 1 of 3

# AUDIT TRAIL 2/3

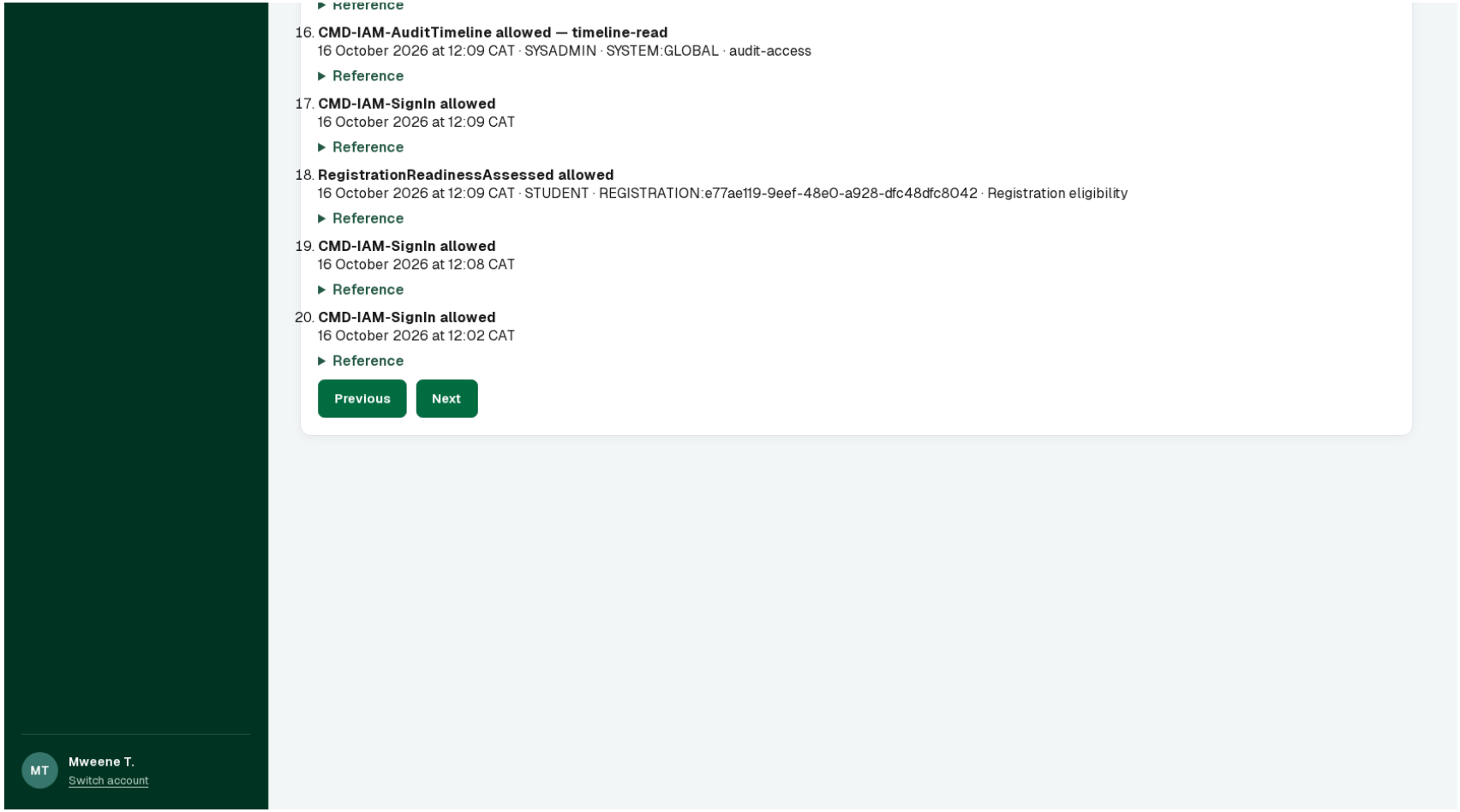


▸ Reference

6. **CoursePlanViewed allowed**
   16 October 2026 at 12:21 CAT · STUDENT · REGISTRATION:c7ec2aaa-8367-4a03-aa05-f25c00a2c14f · Registration eligibility
   ▸ Reference
7. **CMD-IAM-SignIn allowed**
   16 October 2026 at 12:21 CAT
   ▸ Reference
8. **ApplicationDraftSaved allowed**
   16 October 2026 at 12:21 CAT · APP · APPLICATION:c089b83f-4ebe-474b-814e-24f40975e71c · Applicant self-service
   ▸ Reference
9. **ApplicationDraftCreated allowed**
   16 October 2026 at 12:21 CAT · APP · APPLICATION:c089b83f-4ebe-474b-814e-24f40975e71c · Applicant self-service
   ▸ Reference
10. **CMD-IAM-SignIn allowed**
    16 October 2026 at 12:21 CAT
    ▸ Reference
11. **CMD-IAM-SignIn allowed**
    16 October 2026 at 12:13 CAT
    ▸ Reference
12. **CoursePlanViewed allowed**
    16 October 2026 at 12:13 CAT · STUDENT · REGISTRATION:e77ae119-9eef-48e0-a928-dfc48dfc8042 · Registration eligibility
    ▸ Reference
13. **CoursePlanViewed allowed**
    16 October 2026 at 12:13 CAT · STUDENT · REGISTRATION:e77ae119-9eef-48e0-a928-dfc48dfc8042 · Registration eligibility
    ▸ Reference
14. **RegistrationReadinessAssessed allowed**
    16 October 2026 at 12:13 CAT · STUDENT · REGISTRATION:e77ae119-9eef-48e0-a928-dfc48dfc8042 · Registration eligibility
    ▸ Reference
15. **CMD-IAM-SignIn allowed**
    16 October 2026 at 12:13 CAT
    ▸ Reference
16. **CMD-IAM-AuditTimeline allowed — timeline-read**
    16 October 2026 at 12:09 CAT · SYSADMIN · SYSTEM:GLOBAL · audit-access
    ▸ Reference

Reference appendix · full-width page continuation 2 of 3

# AUDIT TRAIL 3/3




Reference appendix · full-width page continuation 3 of 3